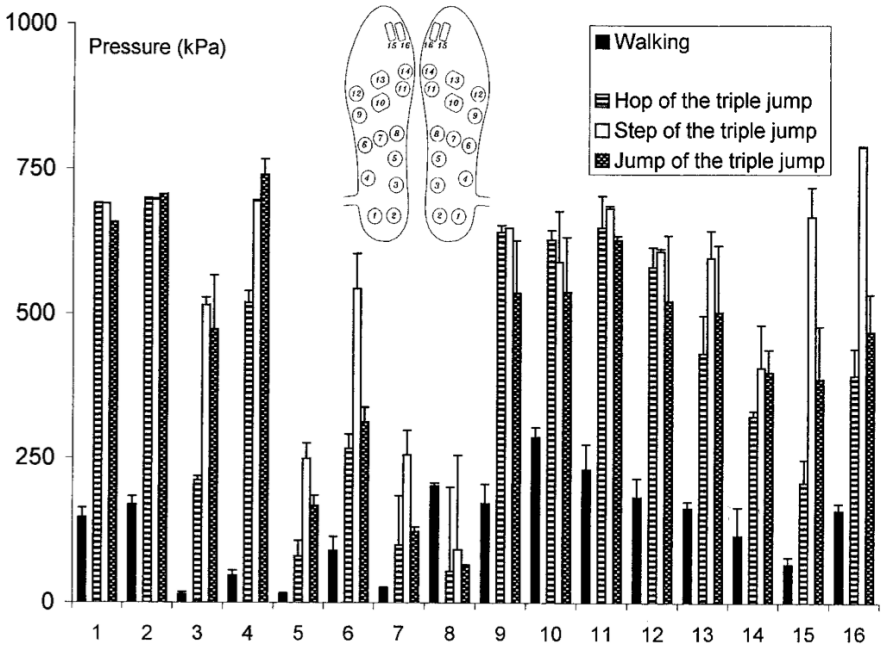


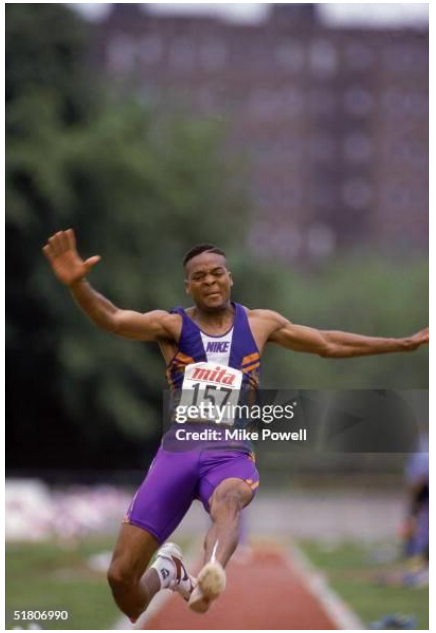
Modul 5.3 Bevægeapparatet

Rogerio Hirata – d.5 Nov 2026

AALBORG UNIVERSITY







2. Example of the peak pressures of the various sensors for one male jumper measured during the hop, step and jump phases e triple jump and during walking. The missing standard deviation bars indicate that the signals exceeded the range of ensors.

Measured, external: about 15 × BW vertical GRF on one leg in the step (15.2 ± 3.3 BW, ≈10.6 kN, 0.16 s contact). This is the mean peak from 7 national-level jumpers (Perttunen et al., 2000). The highest single value was 21.6 BW (Heinonen et al., 2001).

Modul 5.3 Bevægeapparatet

Underekstremitetens (LE) – systematiske anatomi

Lower extremity – systematic anatomy

Sådan læser du slides

Normal tekst = dansk

Kursiv tekst = latinsk anatomisk term

Kursiv, mindre og grøn = engelsk

Tekstbøger: Bevægeapparatets anatomi og Sobotta Atlas of Anatomy

Department of Health Science and Technology, Aalborg University

Systematisk anatomi

Systematic anatomy

Systematisk anatomi giver den grundlæggende 'delliste', før vi studerer regionale relationer.
Systematic anatomy gives the basic 'parts list' before we study regional relationships.

Knogler og landemærker
Bones and anatomical landmarks

→

Led og ligamenter
Joints and ligaments

→

Muskler
Muscles

Knoglerne er rammen og giver palpable landemærker
Bones provide the framework and palpable landmarks

Led og ligamenter definerer bevægelse og stabilitet
Joints and ligaments define movement and stability

Musklerne skaber bevægelse og stabiliserer aktivt benet
Muscles create movement and actively stabilize the lower limb

Ræsonnement: Underekstremiteten er bygget til stabilitet og vægtbæring først, mobilitet bagefter – det modsatte af overekstremiteten.
The lower limb is built for stability and weight bearing first, mobility second – the opposite of the upper limb.

Hvad kan din underekstremitet gøre?

What can your lower limb do?

Bære kroppens vægt
Bear body weight

Stående og siddende
Standing and sitting

Hofte, knæ og ankel i linje
Hip, knee and ankle aligned

Gå og løbe
Walk and run

Gangcyklus: stance og swing
Gait cycle: stance and swing

Kræver koordination af alle led
Needs all joints working together

Holde balancen
Keep balance

Stå på ét ben
Single-leg stance

Bækkenet holdes vandret
The pelvis is held level

Rejse sig og hoppe
Rise and jump

Hofte- og knæekstension
Hip and knee extension

Plantarfleksion giver afsæt
Plantar flexion gives push-off

Kæden hofte – knæ – ankel – fod skal fungere som en enhed.
The chain hip – knee – ankle – foot must work as one unit.

Svigt i ét led giver kompensation i de andre.
Failure in one joint causes compensation in the others.



Læringsmål

Learning objectives

1. **Redegøre for de vigtigste knogler og anatomiske landemærker i underekstremiteten**
   Describe the major bones and anatomical landmarks of the lower extremity

2. **Redegøre for de vigtigste led og ligamenter (hofte, knæ, ankel og fod) samt forklare deres betydning for bevægelse og stabilitet**
   Describe the major joints and ligaments (hip, knee, ankle and foot) and explain their role in movement and stability

3. **Redegøre for de vigtigste muskelgrupper omkring hofte og balle samt forklare deres rolle i bevægelse og stabilisering af bækkenet**
   Describe the major muscle groups of the hip and gluteal region and explain their role in movement and pelvic stabilization

4. **Redegøre for de vigtigste muskelgrupper i lår og underben og relatere deres placering til deres funktion**
   Describe the major muscle groups of the thigh and leg and relate their position to their function

5. **Redegøre for de vigtigste strukturer i foden, herunder fodbuer, plantaraponeurosen og funktionelt vigtige fodmuskler**
   Describe the major structures of the foot, including the arches, plantar aponeurosis and functionally important foot muscles

Underekstremitetens knogler

Bones of the lower extremity

Bækkenbælte
Pelvic girdle

Cingulum pelvicum

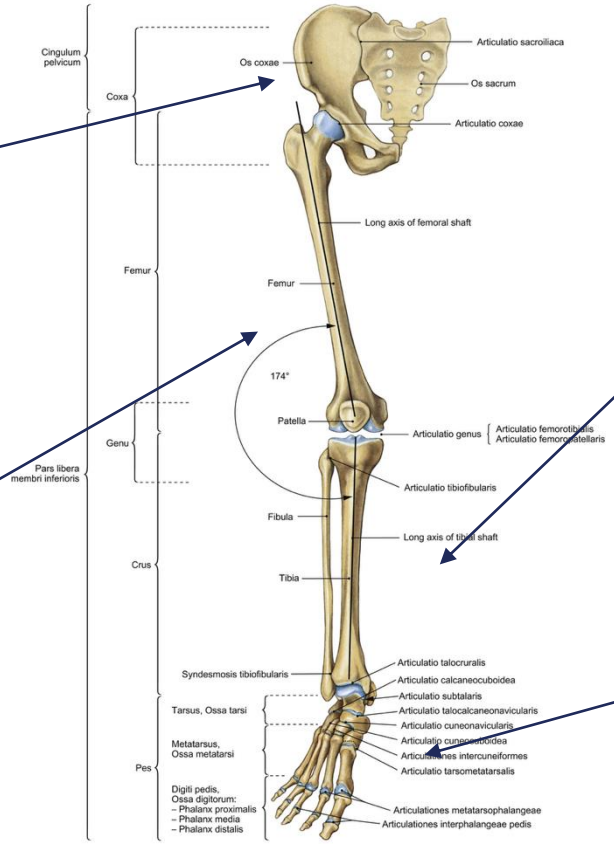
Os coxae: ilium, ischium, pubis

Os sacrum og *os coccygis*

Lår
Thigh

Femur

Patella: Sesamben i quadricepssenen
Sesamoid bone in the quadriceps tendon



Underben
Leg

Tibia: medialt, bærer vægten
Medial, weight bearing

Fibula: lateralt, ankelgaffel og muskelfæste
Lateral, ankle mortise and muscle attachment

Fod
Foot

Ossa tarsi: 7 knogler
Tarsal bones

Ossa metatarsi I–V
Metatarsals

Phalanges: hallux 2, øvrige 3 tæer
Hallux 2, other toes 3

Husk: Fibula bærer kun en lille del af vægten, men er afgørende for ankelleddets stabilitet.
The fibula carries little weight but is essential for ankle stability.
Wang Q, Whittle M, Cunningham J, Kenwright J (1996), *Clinical Orthopaedics and Related Research* 330:261–270 (DOI 10.1097/00003086-199609000-00034; PMID 8804301).

Bækkenbæltet: Os coxae

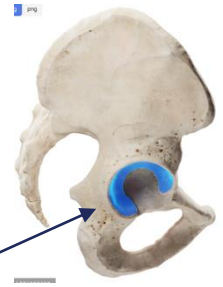
The pelvic girdle

Os ilium Ilium: *Ala ossis ilii*, *crista iliaca*, *SIAS* og *SIPS*
Wing, iliac crest, anterior and posterior superior iliac spines

Os ischii Ischium: *Tuber ischiadicum* og *spina ischiadica*
Ischial tuberosity and spine

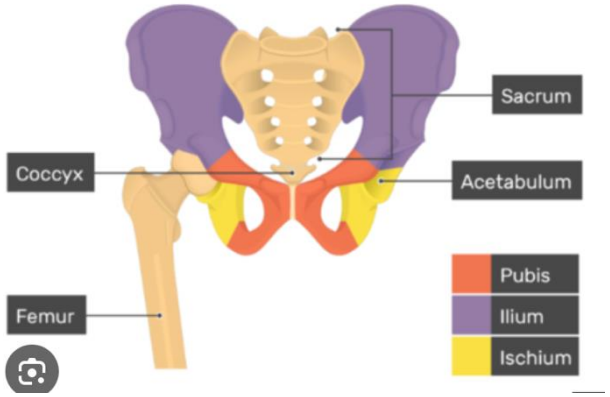
Os pubis Pubis: *Ramus superior* og *inferior*, *tuberculum pubicum*
Superior and inferior ramus, pubic tubercle

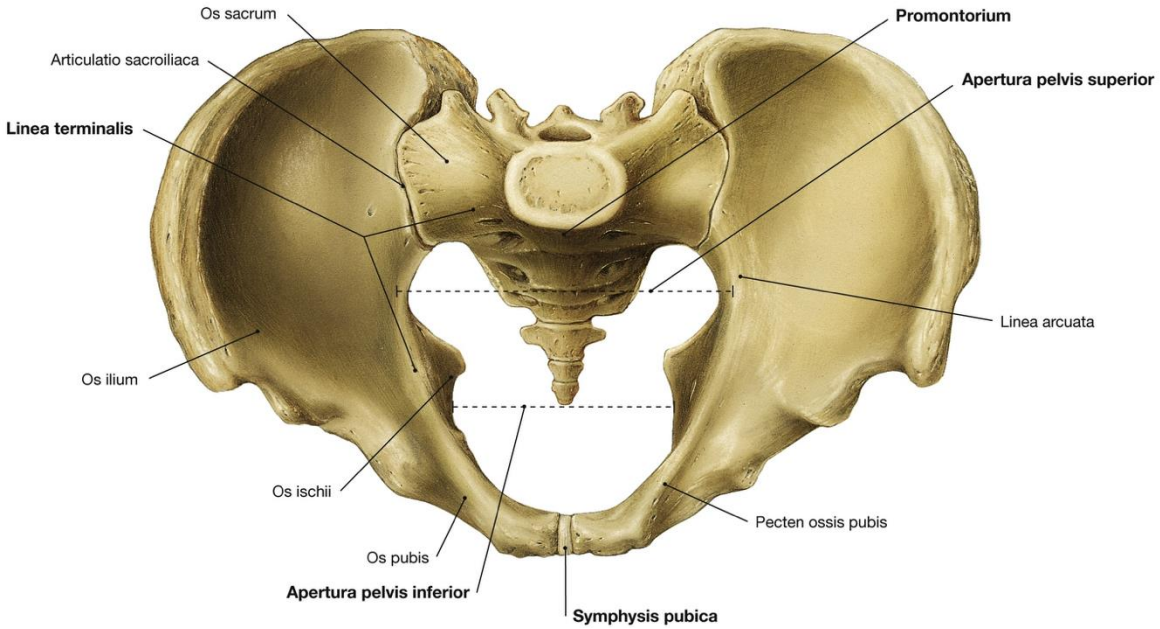
Acetabulum: Ledskålen: *facies lunata* og *fossa acetabuli*
Articular socket: lunate surface and acetabular fossa





Skematisk: de tre knogler mødes i acetabulum





Udvikling: Knoglerne forbener sammen i løbet af teenageårene, tidligere hos piger end hos drenge. (Fabricant PD, et al. J Bone Joint Surg Am 2013;95(3):230-6.)
The boned at pelvic girdle fuses only in the teenage years (Y-shaped cartilage).

Femur

The femur

Proksimalt
Proximal

Caput femoris med *fovea capitis*
Head of femur

Collum femoris
Neck of femur

Trochanter major og *minor*
Greater and lesser trochanter

Linea og *crista intertrochanterica*

Corpus
Shaft

Linea aspera: Muskelfæste til adduktorer og vasti
Attachment for adductors and vasti

Tuberositas glutea: Fæste for m. gluteus maximus
Gluteus maximus attachment

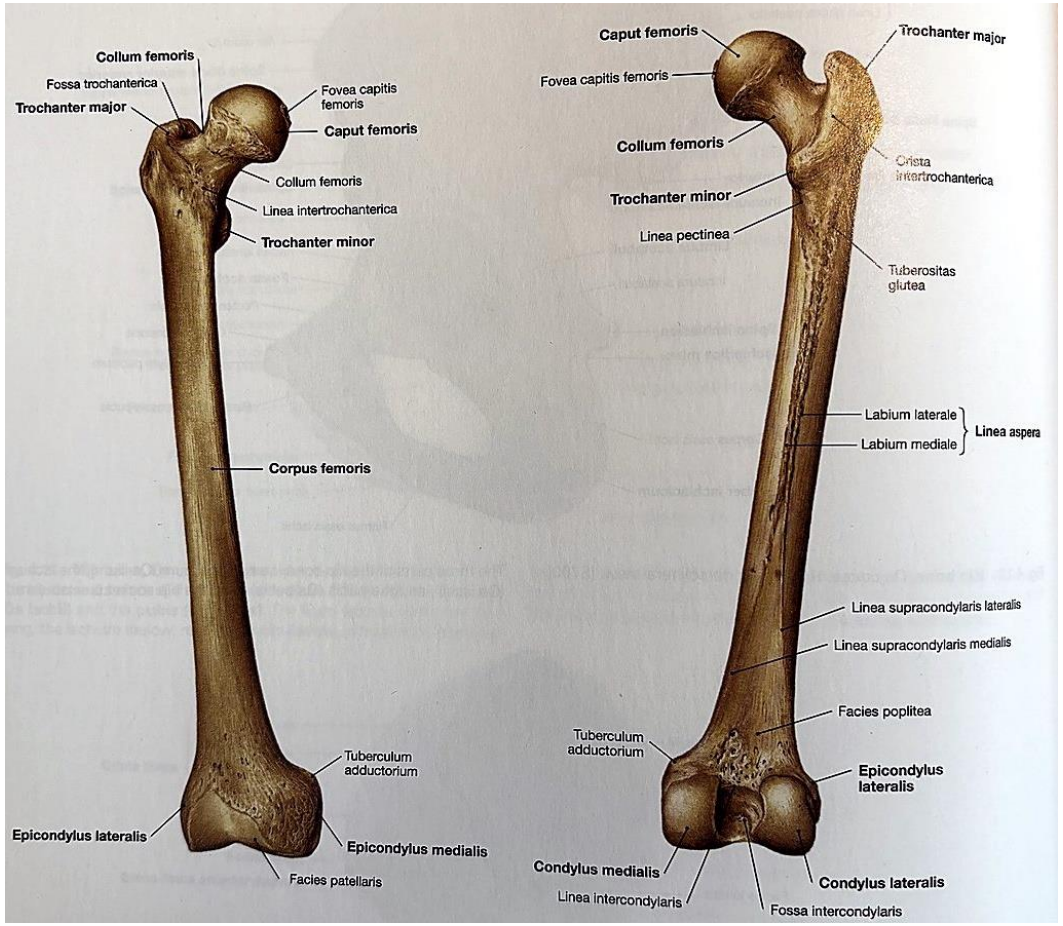
Distalt
Distal

Condylus medialis og *lateralis*

Epicondylus medialis og *lateralis*

Tuberculum adductorium

Facies patellaris: Rille for patella
Groove for the patella



Klinik: Collum femoris-fraktur hos ældre efter lavenergi-fald. Benet er typisk forkortet og udadroteret.
Femoral neck fracture in older adults after a low-energy fall. The leg is typically shortened and externally rotated.

Patella, tibia og fibula

Patella, tibia and fibula

Patella

Kneecap

Sesamben i *quadricepssenen*

Sesamoid bone in the quadriceps tendon

Basis og *apex patellae*

Øger quadriceps' løftearm

Increases the lever arm of the quadriceps

Tibia

Shinbone

Condylus medialis og *lateralis*

Eminentia intercondylaris

Tuberositas tibiae: Fæste for lig. patellae

Patellar ligament attachment

Margo anterior: 'skinnebenet'

Malleolus medialis

Fibula

Calf bone

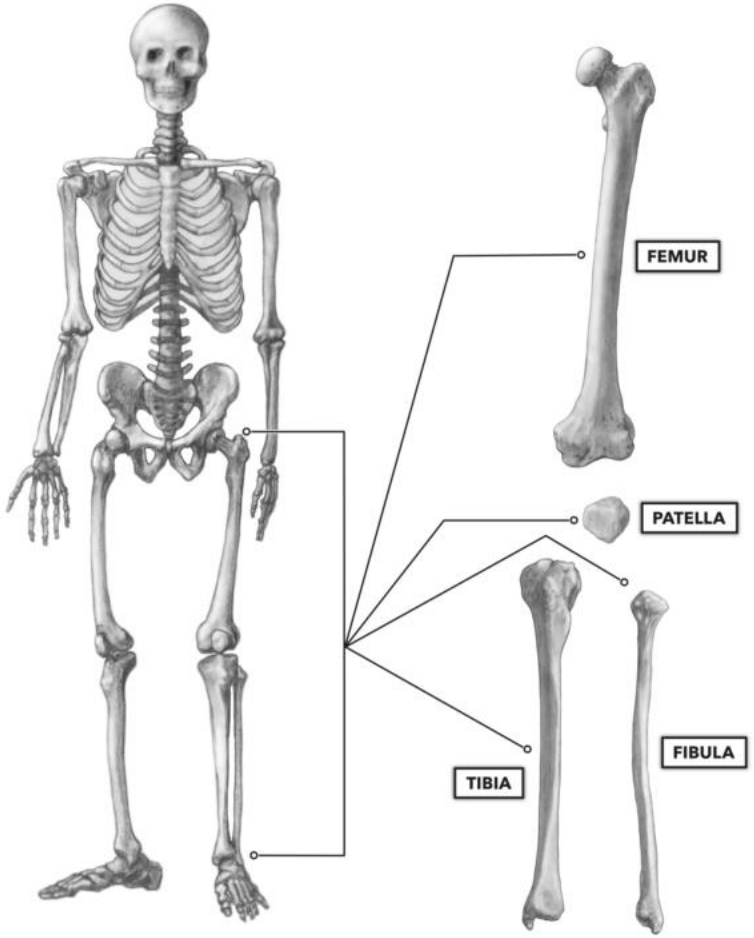
Caput fibulae

Collum fibulae

Malleolus lateralis: Når længere distalt end den mediale

Extends more distally than the medial malleolus

Membrana interossea cruris



Klinik: Ved knæluksation (tibiofemoral) forskydes tibia i forhold til femur, og de store ledbånd brister. Kollateralbåndets fibulære fæste og n. fibularis communis ved collum fibulae er udsat, og a. poplitea ligger tæt på bagsiden af knæet. Skades nerven, giver det dropfod. Må ikke forveksles med patellaluksation..

In knee dislocation (tibiofemoral), the tibia is displaced relative to the femur and major ligaments rupture. The fibular attachment of the lateral collateral ligament and the common fibular nerve at the fibular neck are exposed, and the popliteal artery lies close behind the knee. Nerve injury causes foot drop. It should not be confused with patellar dislocation.

Randall ZD, Strok MJ, Mazzola JW, Agrawal R, Yaeger LH, Berkes MB. The known and unknown reality of knee dislocations: a systematic review. Injury 2024;55(11):111904.

Patella

Kneecap

Patella

Kneecap

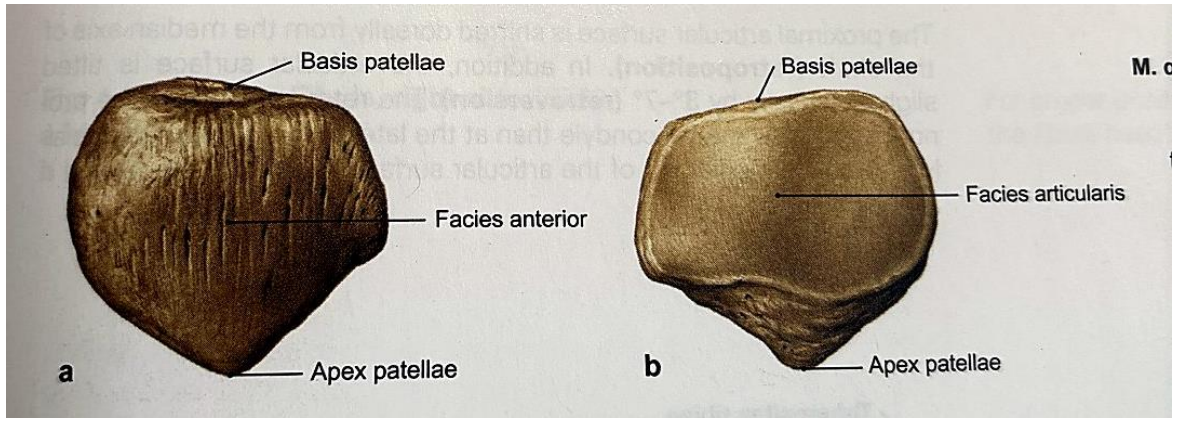
Sesamben i *quadricepssenen*

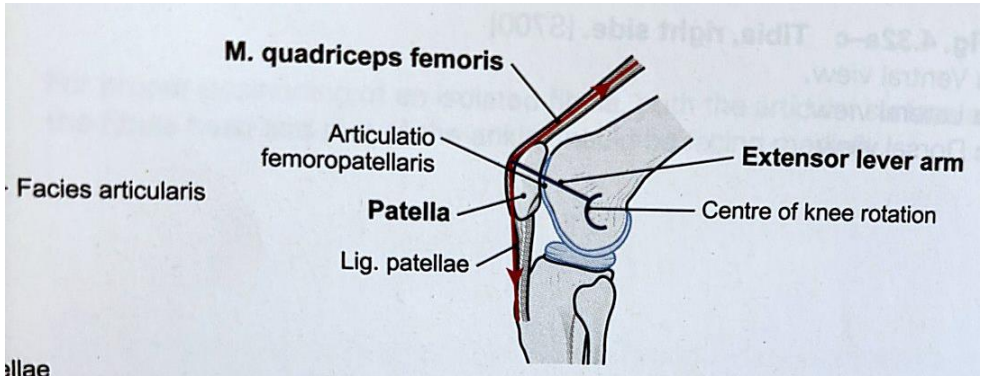
Sesamoid bone in the quadriceps tendon

Basis og *apex patellae*

Øger quadriceps' løftearm

Increases the lever arm of the quadriceps







Klinik: Patella bipartita skyldes ufuldstændig forbening af patella, så et fragment (oftest superolateralt) forbliver forbundet med en fibrøs/bruskagtig synkondrose. Den er ofte et tilfældigt fund, men kan give smerter, især hos unge idrætsudøvere.

A bipartite patella results from incomplete ossification, leaving a fragment (most often superolateral) joined by a fibrocartilaginous synchondrosis. It is often an incidental finding but can cause pain, especially in young athletes.

Kallini J, Micheli LJ, Miller PE, Kramer DE, Kocher MS, Heyworth BE. Operative treatment of bipartite patella in pediatric and adolescent athletes: a retrospective comparison with a nonoperatively treated cohort. Orthop J Sports Med 2021;9(1):2325967120967125

Tibia

tibia

Tibia

Shinbone

Condylus medialis og *lateralis*

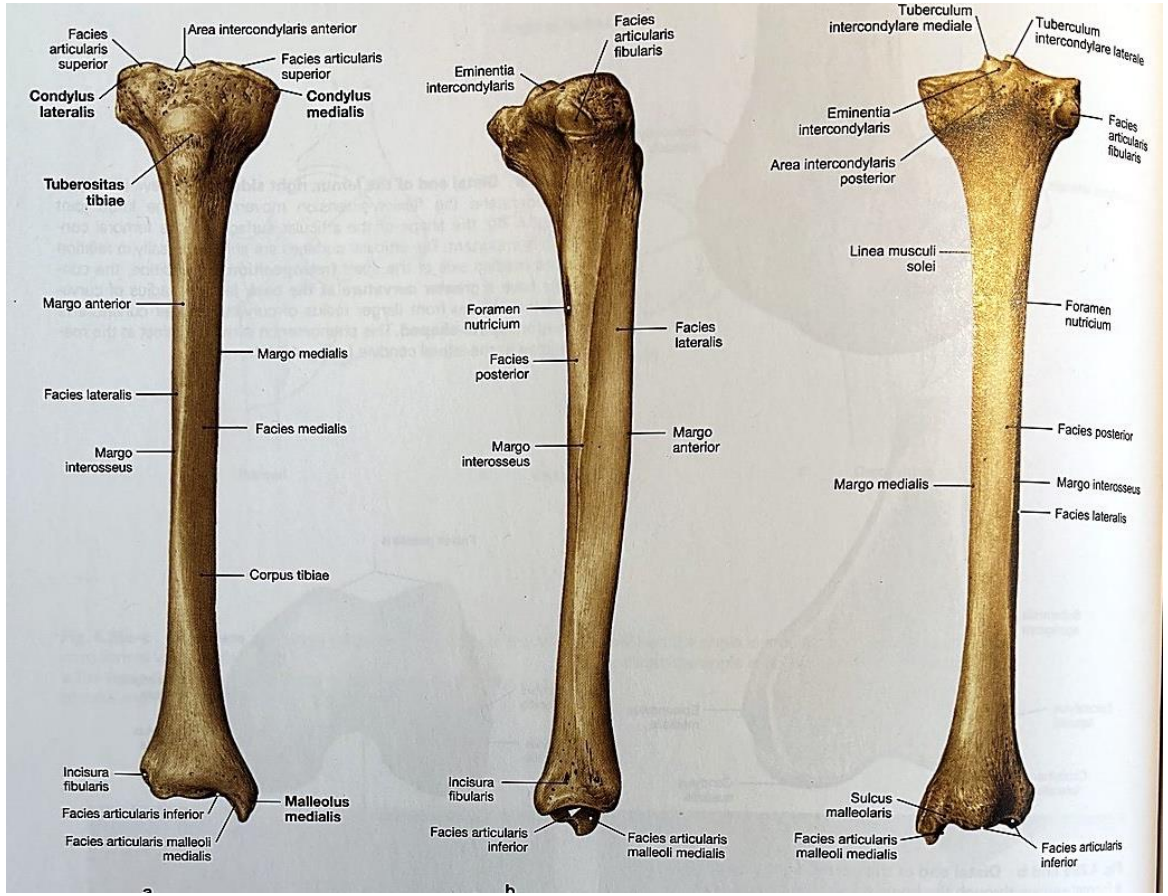
Eminentia intercondylaris

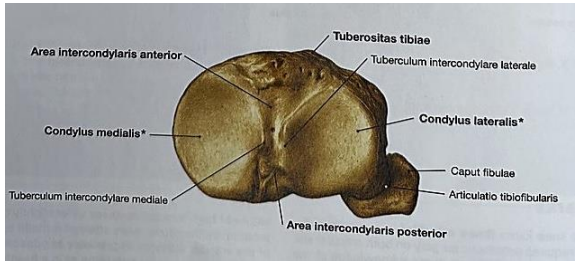
Tuberositas tibiae: Fæste for lig. patellae

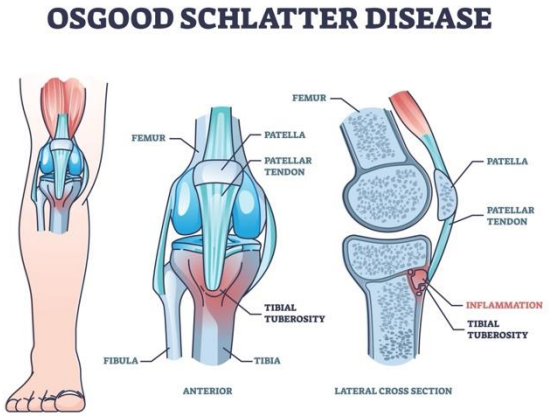
Patellar ligament attachment

Margo anterior: 'skinnebenet'

Malleolus medialis







Klinik: Tuberositas tibiae er fæstet for lig. patellae. Hos unge i vækst kan gentagen træk fra quadriceps give apofysitis (Osgood-Schlatter) med smerter og en synlig knude. Tidligere regnet som godartet, men en dansk registerundersøgelse viser, at mange voksne stadig har besvær.

The tibial tuberosity is the insertion of the patellar ligament. In growing adolescents, repeated quadriceps traction can cause apophysitis (Osgood-Schlatter disease) with pain and a visible prominence. It was long considered benign, but a Danish registry study shows that many adults still have problems.

Krommes K, Bjerre A, Thorborg K, Nielsen MF, Hölmich P. Long-term knee health in adults with a history of adolescent Osgood-Schlatter: a national cohort study of patients in secondary care in Denmark 1977-2020. *Sports Med* 2025;55(7):1769-81.

Fibula

Fibula

Fibula

Calf bone

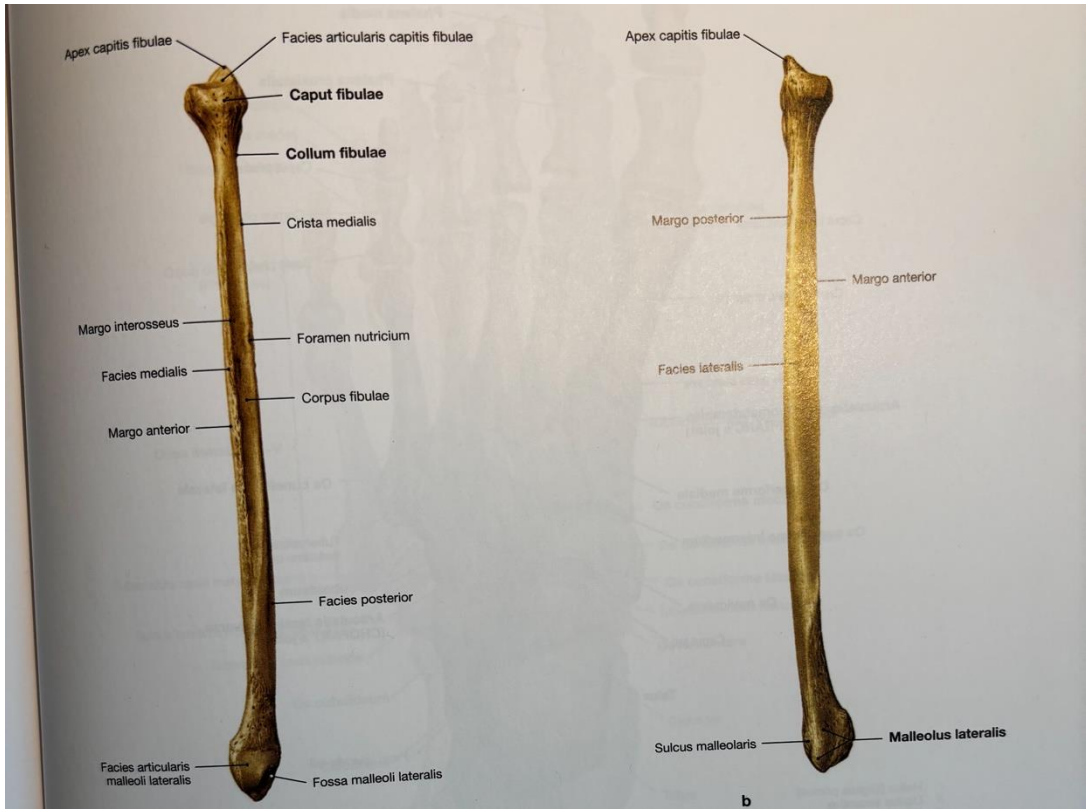
Caput fibulae

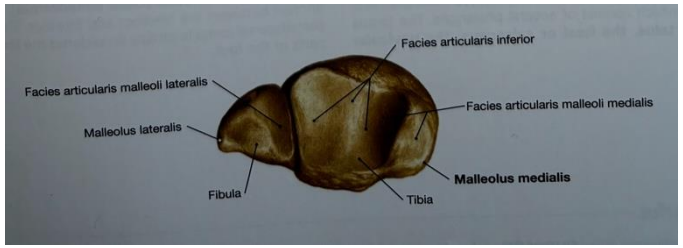
Collum fibulae

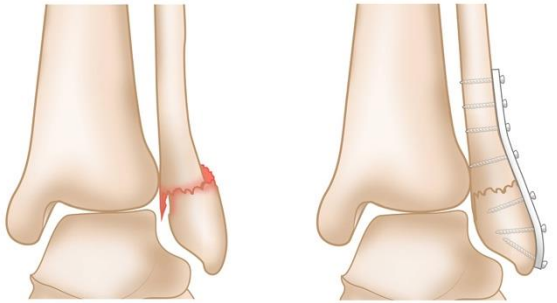
Malleolus lateralis: Når længere distalt end den mediale

Extends more distally than the medial malleolus

Membrana interossea cruris







Klinik: Collum fibulae er tæt på n. fibularis communis. Fraktur eller tryk her kan give dropfod.

The fibular neck lies close to the common fibular nerve. Fracture or pressure here can cause foot drop.

Klinik: Fraktur af malleolus lateralis er den hyppigste ankelfraktur (ca. 55 %) og opstår typisk ved vridtraume af anklen.

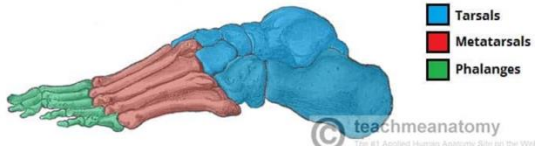
A fracture of the lateral malleolus is the most common ankle fracture (approx. 55%) and typically occurs as a result of a twisting injury to the ankle.

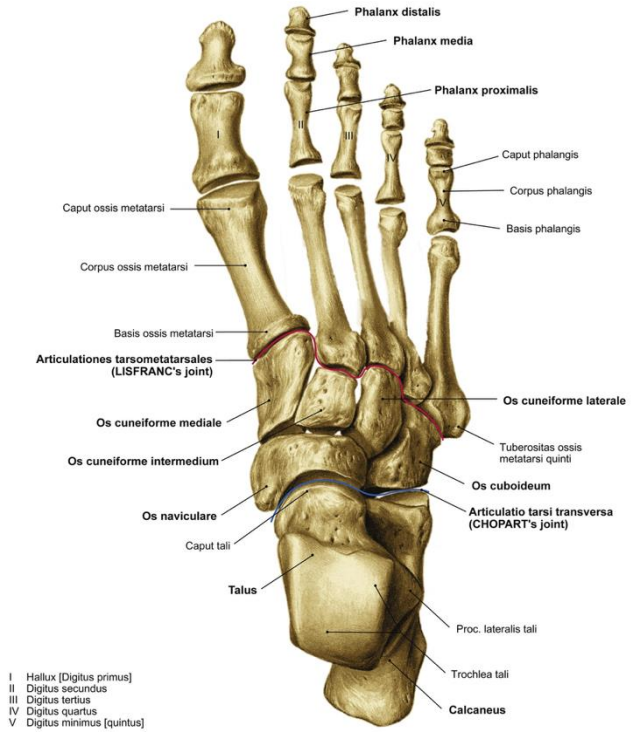
"The predominant mode of injury was falls (61%) followed by sports (22%)."

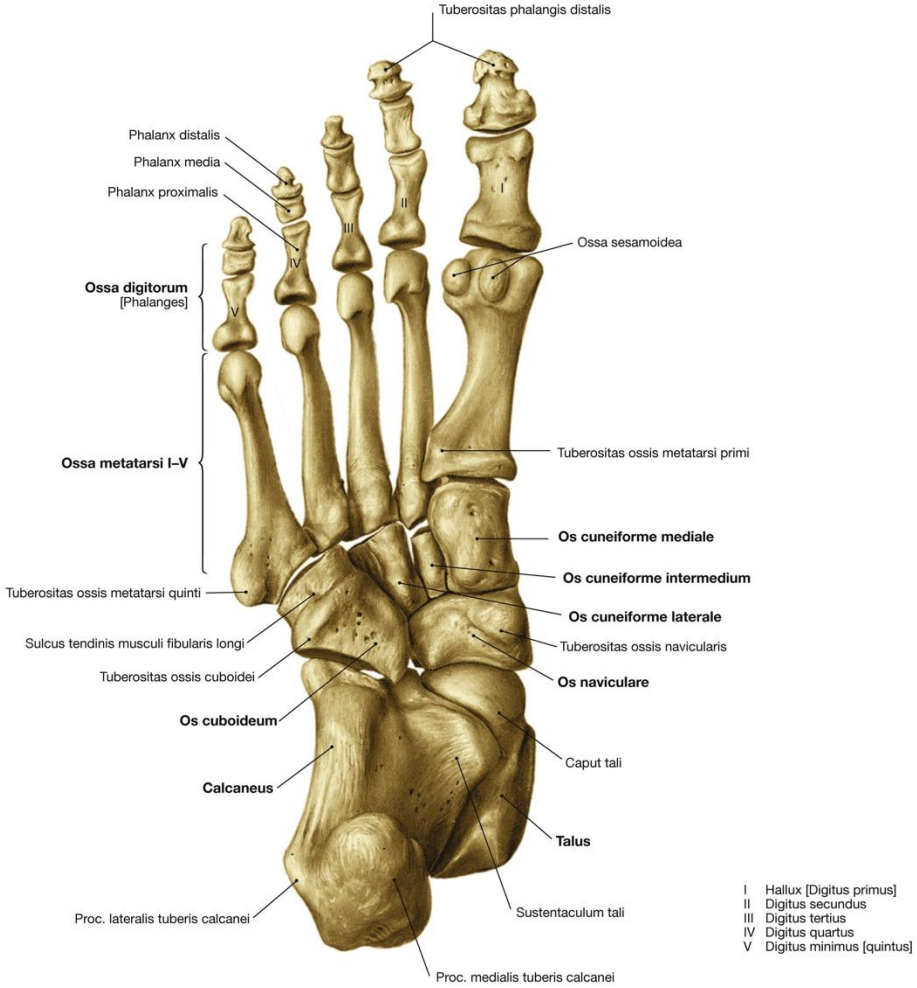
Elsoe, Ostgaard & Larsen, "Population-based epidemiology of 9767 ankle fractures", Foot Ankle Surg 24(1):34-39.

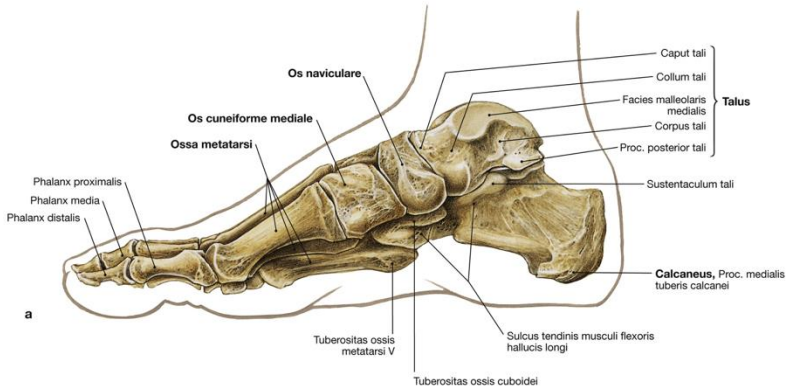
Fodens knogler

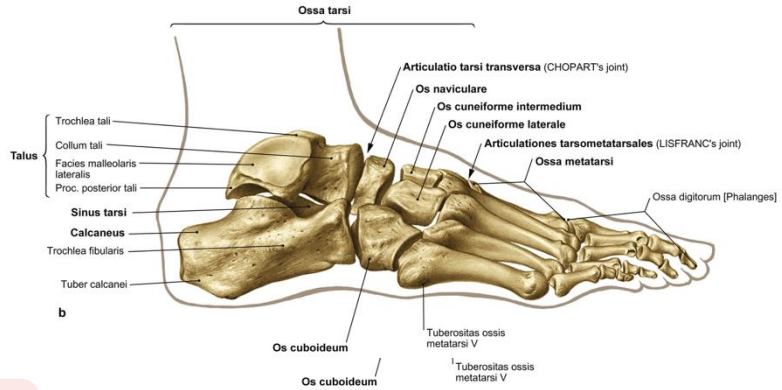
Bones of the foot











Klinik: Hælbenbrud (calcaneal frakturer) er de mest almindelige brud på fodrodens knogler (tarsalknoglerne) og udgør 1 % til 2 % af alle knoglebrud.

Calcaneal fractures are the most common fracture of the tarsal bones and represent 1%-2% of all fractures.

Buckley RE, Tough S. Displaced intra-articular calcaneal fractures. J Am Acad Orthop Surg. 2004;12(3):172–178

Fodens knogler

Bones of the foot

Baghold
Hindfoot

Talus og *calcaneus*

Calcaneus: *tuber calcanei* (hælen) og *sustentaculum tali*
Heel bone: calcaneal tuberosity and sustentaculum tali

Talus: *caput*, *collum* og *trochlea tali* i ankelleddet
Head, neck and trochlea in the ankle joint

Tuberositas ossis metatarsalis quinti (V): palpabel, fæste for m. fibularis brevis
Palpable, fibularis brevis attachment



Midtfod
Midfoot

Os naviculare, *os cuboideum*, 3 *ossa cuneiformia*

Huskeregel til tarsus
Mnemonic for the tarsal bones

Tiger **C**ubs **N**eed **MILC**

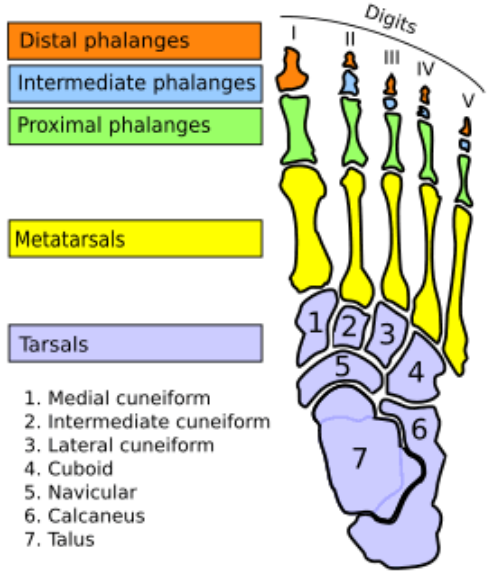
Talus, **C**alcaneus, **N**avicular, **M**edial, **I**ntermediate og **L**ateral cuneiform, **C**uboid

Klinik: Accessorisk navicular (sekundært forbeningscenter ved tuberositas ossis navicularis) kan give smerter medialt og ses ofte sammen med fleksibel fladfod.



Forfod
Forefoot

5 *ossa metatarsi* og *phalanges*



Læringsmål

Learning objectives

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   Describe the major bones and anatomical landmarks of the lower extremity

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   Describe the major joints and ligaments (hip, knee, ankle and foot) and explain their role in movement and stability

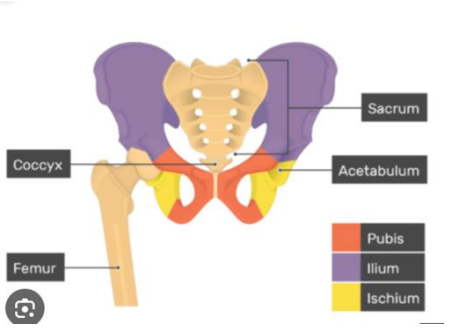
3. Redegøre for de vigtigste muskelgrupper omkring hofte og balle samt forklare deres rolle i bevægelse og stabilisering af bækkenet
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   Describe the major structures of the foot, including the arches, plantar aponeurosis and functionally important foot muscles

Hofteleddet

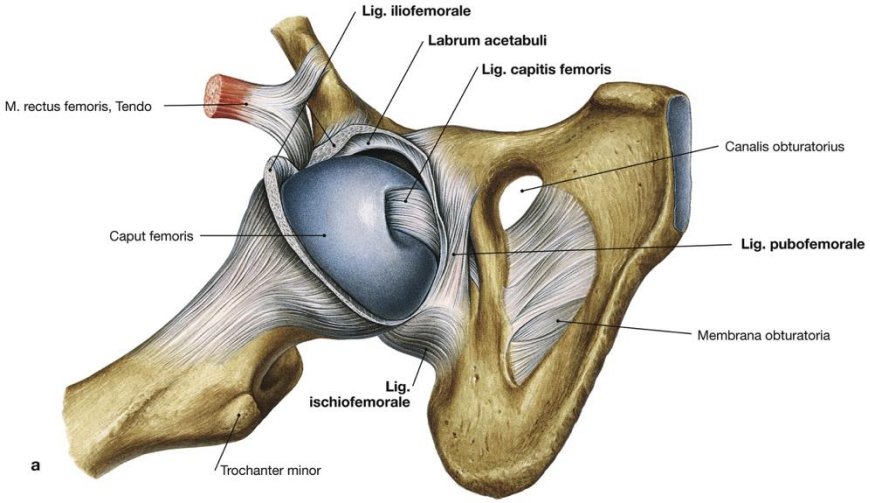
The hip joint

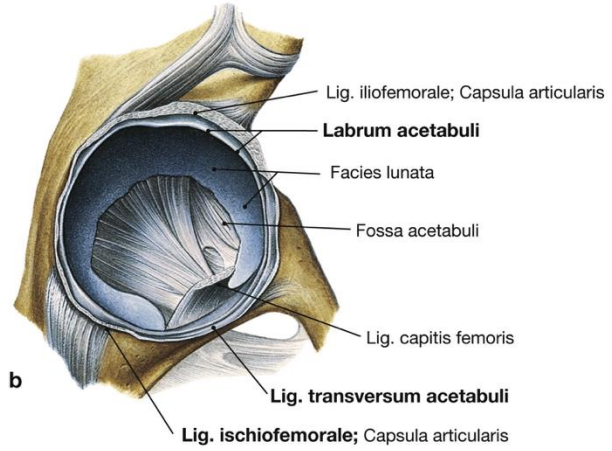


Articulatio coxae: kugleled
Ball-and-socket joint

Caput femoris i *acetabulum*, forstærket af *labrum acetabulare*
Femoral head in the acetabulum, deepened by the labrum

Stor knoglekongruens, stærk kapsel og stærke ligamenter
High bony congruence, strong capsule and ligaments





Omtrentligt bevægeudslag
Approximate range of motion

- Fleksion – ekstension: ca. 130° – 10°
- Abduktion – adduktion: ca. 40° – 30°
- Udadrotation – indadrotation: ca. 40° – 50°

Skulder: mobilitet
Shoulder: mobility

vs.

Hofte: stabilitet
Hip: stability

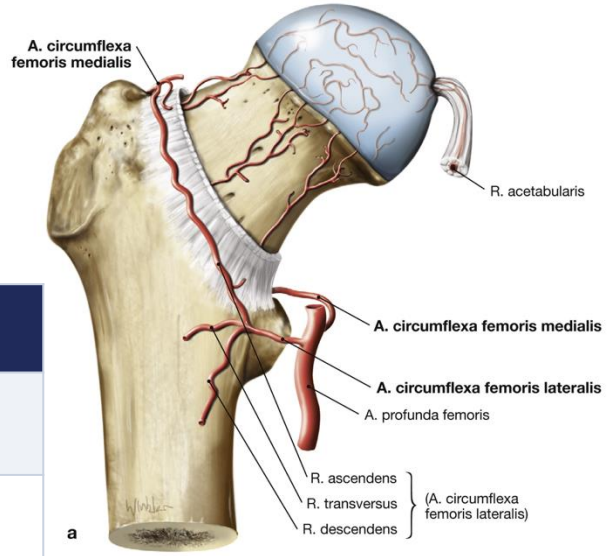
Luksation er sjælden og kræver høj energi.
Dislocation is rare and requires high energy.

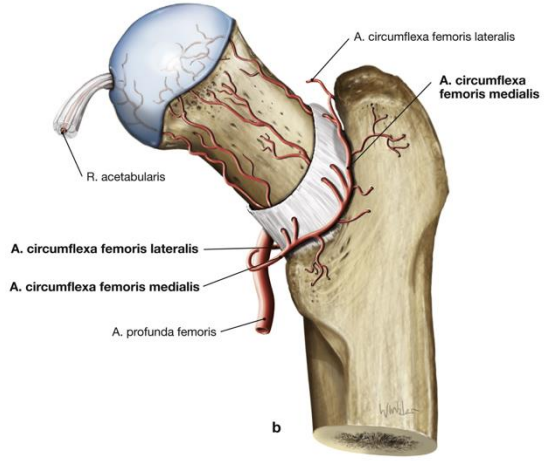
Hoftens ligamenter og blodforsyning

Hip ligaments and blood supply

Ligament	Forløb	Begrænser
Lig. iliofemorale	*Spina iliaca anterior inferior* og acetabulum til *linea intertrochanterica*; Y-formet	Ekstension (stærkest)
Lig. pubofemorale	*Ramus superior ossis pubis* til collum	Abduktion og ekstension
Lig. ischiofemorale	*Os ischii* til collum	Indadrotation og ekstension
Lig. capitis femoris	*Fovea capitis* til *incisura acetabuli* og *lig. transversum acetabuli*	

↳ Fører lille arterie til caput (kar-function usikker hos voksene)





Blodforsyning til caput femoris

Blood supply to the femoral head

Primært *a. circumflexa femoris medialis*

Mainly the medial circumflex femoral artery

Retinaculære grene løber langs collum op til caput

Retinacular branches run along the neck to the head

Lille bidrag fra *a. ligamenti capitis*

Minor contribution in adults

Klinik: Intrakapsulær collum femoris-fraktur kan afbryde retinaculære kar og give avaskulær nekrose af caputs knogle.

An intracapsular femoral neck fracture can disrupt the retinacular vessels and cause avascular necrosis of the femoral head.

Knæleddet

The knee joint

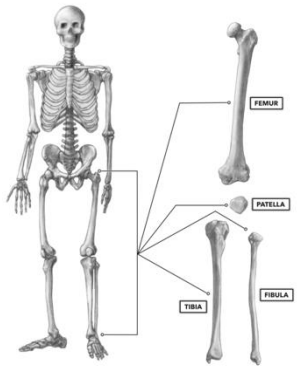
To led i ét

Two joints in one

Articulatio femorotibialis
Tibiofemoral joint

Articulatio femoropatellaris
Patellofemoral joint

Fælles ledhule
Shared joint cavity



Bevægelser

Movements

Fleksion ca. 140°, ekstension 5°
Flexion about 135–150°, extension 0°

Rotation kun når knæet er bøjet. Lat 30°, Med 10°.
Rotation only in flexion

Screw-home: låsning i ekstension
Locking in terminal extension

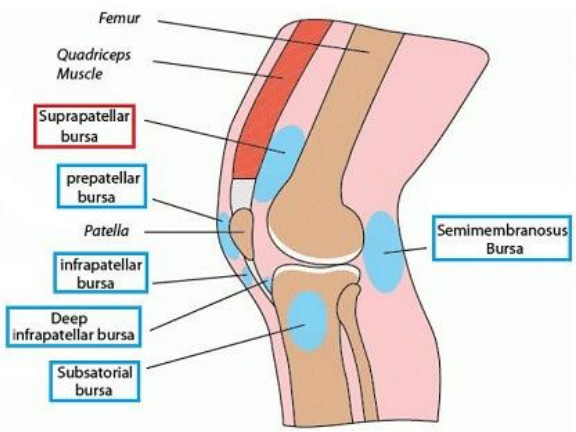
Bursae

Bursae

Bursa suprapatellaris: kommunikerer med ledhulen
Communicates with the joint cavity

Bursa prepatellaris og *infrapatellaris*

Bursa anserina



Lav knoglekongruens
Low bony congruence

→

Afhænger af bløddele
Depends on soft tissue

→

Hyppige skader
Frequent injuries

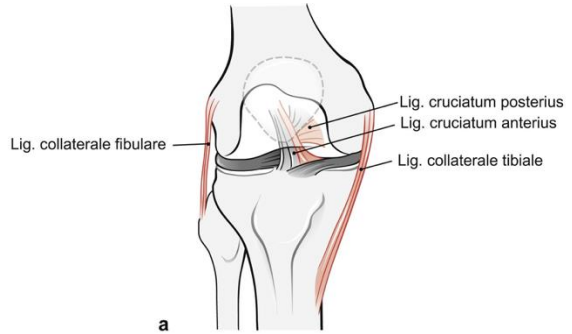
Klinik: Fabella er et sesambeen i m. gastrocnemius' laterale hoved og findes hos mange; hyppigheden ser ud til at være steget over det sidste århundrede. Den kan give posterolateral knæsmerte (fabella-syndrom).
The fabella is a sesamoid bone in the lateral head of gastrocnemius and is present in many people; its prevalence appears to have increased over the past century. It can cause posterolateral knee pain (fabella syndrome). *Berthaume MA, et al. J Anat 2019;235(1):67-79.*

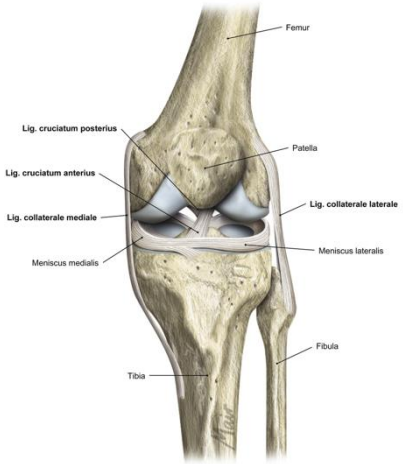


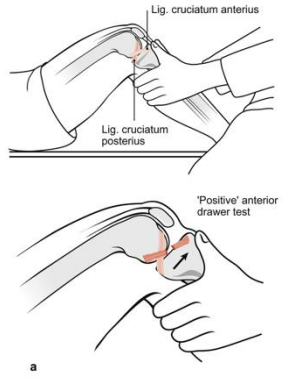
Knæets ligamenter og menisker

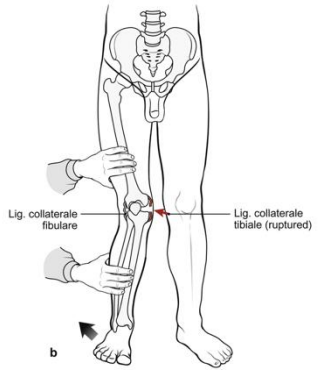
Knee ligaments and menisci

Struktur *Structure*	Forhindrer	Klinisk test
Lig. cruciatum anterius **(forreste korsbånd, ACL)**	Forreste forskydning af tibia	Lachman, forreste skuffetest
Lig. cruciatum posterius **(bageste korsbånd, PCL)**	Bageste forskydning af tibia	Bageste skuffetest
Lig. collaterale tibiale **(mediale sideled, MCL)**	Valgus (indadgående knæ)	Valgusstress
Lig. collaterale fibulare **(laterale sideled, LCL)**	Varus (udadgående knæ)	Varusstress
Meniscus medialis **og** ***lateralis***	Fordeler last, øger kongruens, støtter stabilitet	McMurray, ledlinjesmerte







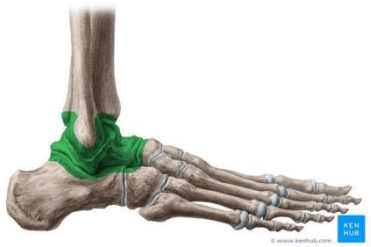


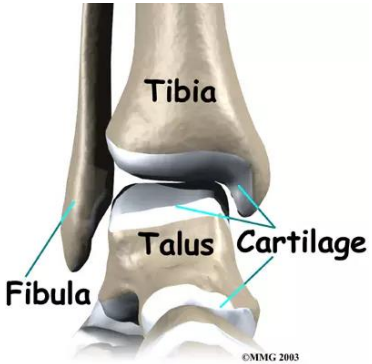
Klinik: Vridtraume med 'knæk' og hurtig hævelse (haemarthros) giver mistanke om ACL-skade. Menisker er kun vaskulariseret perifert og heler dårligt centralt.

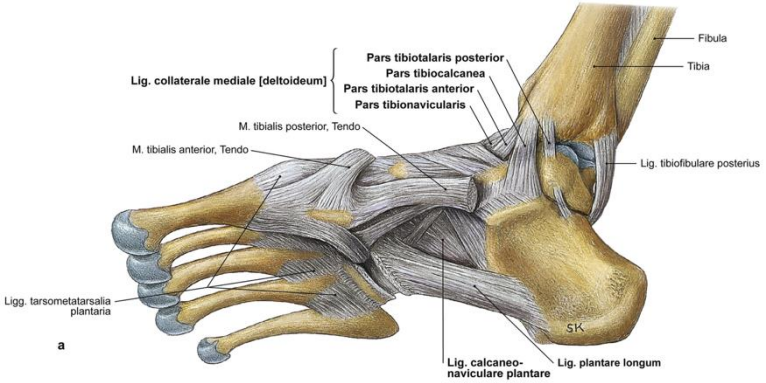
A twisting injury with a 'pop' and rapid swelling (haemarthrosis) suggests ACL injury. The menisci are vascularised only peripherally and heal poorly centrally.

Ankelleddet

The ankle joint



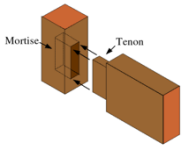


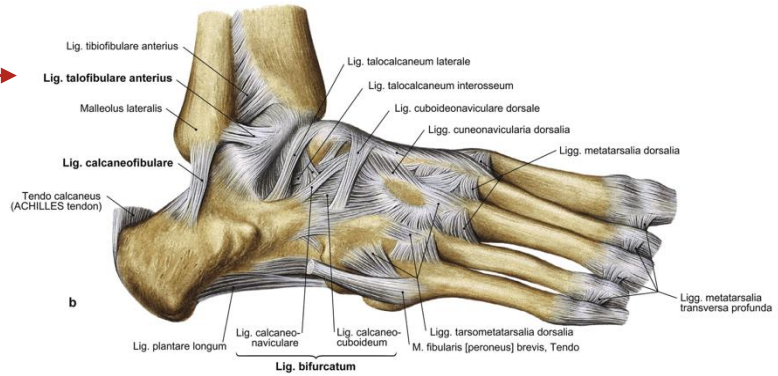


Articulatio talocruralis: hængselled
Hinge joint

Ankelgaffel: *tibia* + begge *malleoler* omslutter *trochlea tali*
The mortise: tibia and both malleoli grip the talar trochlea

Dorsalfleksion ca. 20°, plantarfleksion ca. 40–50°
Dorsiflexion about 20°, plantar flexion about 40–50°





Laterale ligamenter

Lateral ligaments

Lig. talofibulare anterius (ATFL): svagest, oftest skadet
Weakest, most often injured

Lig. Calcaneofibulare (CF)

Lig. talofibulare posterius

Mediale ligament

Medial ligament

Lig. deltoideum: stærkt, firdelt
Strong, four-part deltoid ligament

Skades sjældnere end de laterale
Injured less often than the lateral ligaments



Klinik: Typisk forstuvning: plantarfleksion + inversion. ATFL og CF kan rives.
Typical sprain: plantar flexion plus inversion, tearing ATFL and CF can occur.

Fodens led og buer

Joints and arches of the foot

Fodens led

Foot joints

Articulatio subtalaris: inversion og eversion

Inversion and eversion

Articulatio tarsi transversa (Chopart)

Articulationes tarsometatarsales (Lisfranc)





Tre fodbuer

Three arches

Medial længdebue

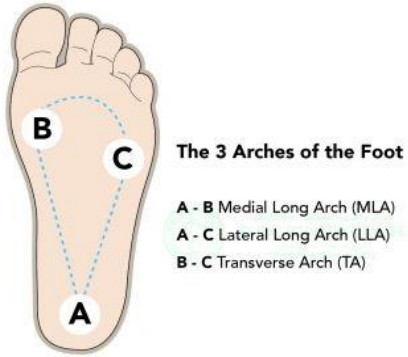
Medial longitudinal arch

Lateral længdebue

Lateral longitudinal arch

Tværbue

Transverse arch



Buens støtte

Arch support

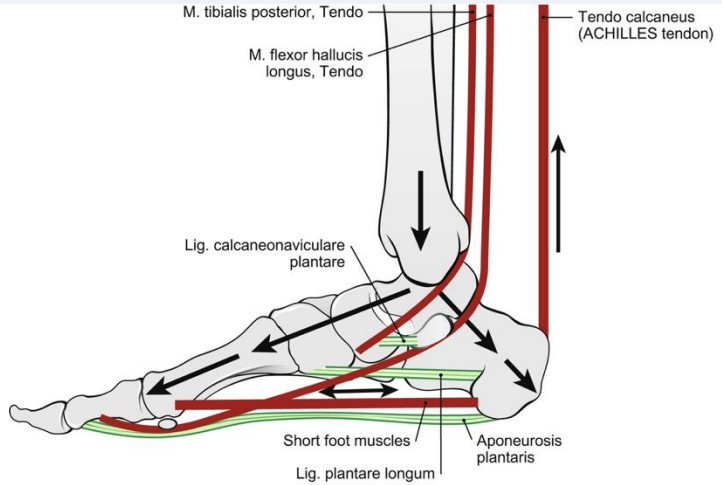
Aponeurosis plantaris

Lig. calcaneonaviculare plantare (spring ligament)

Lig. plantare longum

Sener: tibialis posterior og fibularis longus

Tendons



Klinik: Plantar fasciitis giver hælsmerter ved de første skridt. Fladfod (pes planus) opstår ved svigt af buens støtte, især m. tibialis posterior.

Plantar fasciitis causes heel pain with the first steps. Flat foot (pes planus) follows failure of arch support, especially tibialis posterior.

Læringsmål

Learning objectives

1. Redegøre for de vigtigste knogler og anatomiske landemærker i underekstremiteten
   Describe the major bones and anatomical landmarks of the lower extremity

2. **Redegøre for de vigtigste led og ligamenter (hofte, knæ, ankel og fod) samt forklare deres betydning for bevægelse og stabilitet**
   Describe the major joints and ligaments (hip, knee, ankle and foot) and explain their role in movement and stability

3. Redegøre for de vigtigste muskelgrupper omkring hofte og balle samt forklare deres rolle i bevægelse og stabilisering af bækkenet
   Describe the major muscle groups of the hip and gluteal region and explain their role in movement and pelvic stabilization

4. Redegøre for de vigtigste muskelgrupper i lår og underben og relatere deres placering til deres funktion
   Describe the major muscle groups of the thigh and leg and relate their position to their function

5. Redegøre for de vigtigste strukturer i foden, herunder fodbuer, plantaraponeurosen og funktionelt vigtige fodmuskler
   Describe the major structures of the foot, including the arches, plantar aponeurosis and functionally important foot muscles

Hvad skaber bevægelse?

What generates movement?

Hvis du ved, hvor en muskel ligger, og hvilket led den krydser, kan du ofte forudsige dens funktion.
If you know where a muscle lies and which joint it crosses, you can often predict its function.



Fire eksempler

Four examples

M. rectus femoris: hofte + knæ
Hip flexion and knee extension

M. gastrocnemius: knæ + ankel
Knee flexion and plantar flexion

M. biceps femoris: hofte + knæ
Hip extension and knee flexion

Tovledsmuskler

Two-joint muscles

Kræver forståelse af begge led
Need both joints to be understood

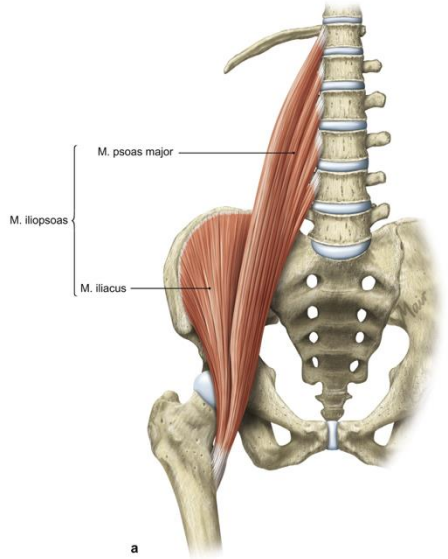
Er mest effektive ved bestemte ledstillinger
Work best at certain joint positions

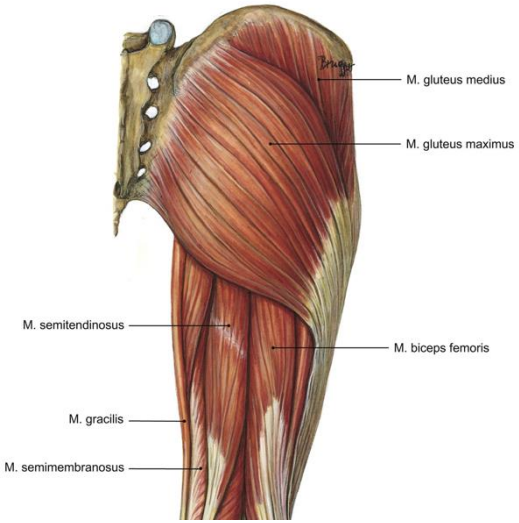
Hofte og balle

Hip and gluteal muscles

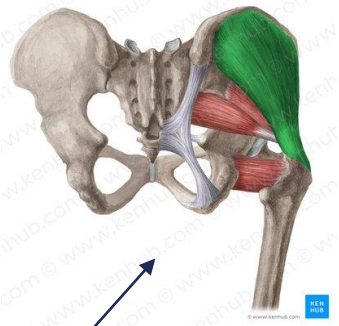
Muskel *Muscle*	**Hovedfunktion**
M. iliopsoas (psoas major og iliacus)	Kraftigste hoftefleksor; stabiliserer lumbalcolumna
M. gluteus maximus	Hofteekstension, udadrotation; rejse sig og trappegang
M. gluteus medius* og *minimus	Abduktion; holder bækkenet vandret i standfasen
M. tensor fasciae latae	Abduktion, fleksion; spænder *tractus iliotibialis*
Dybe laterale rotatorer: *m. piriformis*, *mm. obturatorii*, *mm. gemelli*, *m. quadratus femoris*	Udadrotation; stabiliserer hofteleddet

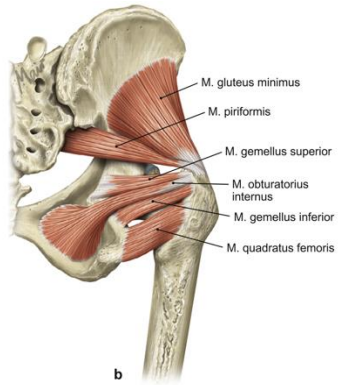
Link: Svækket gluteus medius giver Trendelenburgs tegn (næste slide).
A weak gluteus medius gives a positive Trendelenburg sign (next slide).

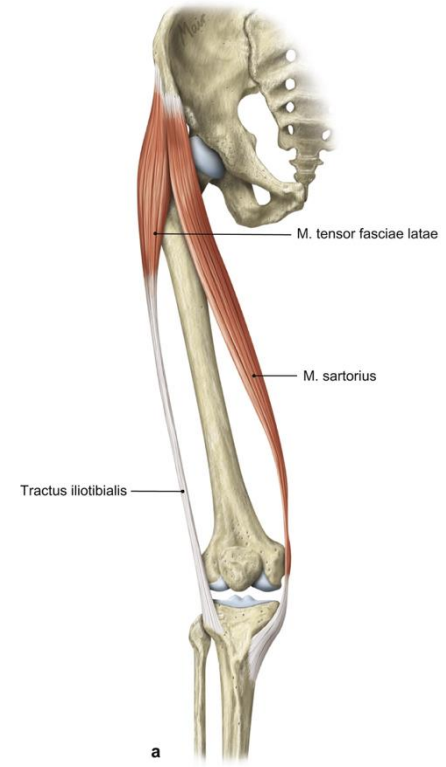




M. gluteus medius





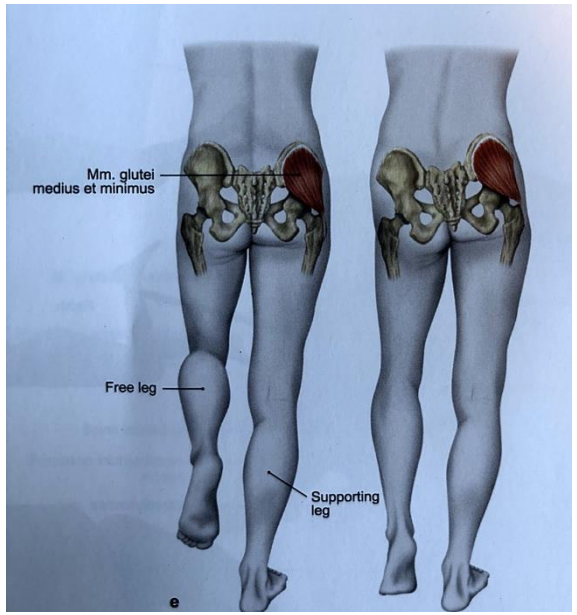


Bækkenstabilitet og Trendelenburgs tegn

Pelvic stability and the Trendelenburg sign

Normal
Normal: pelvis held level

Stable ←



→ Unstable

Svag m. gluteus medius
Weak gluteus medius: pelvis drops on the unsupported side

Standfasen
Stance phase

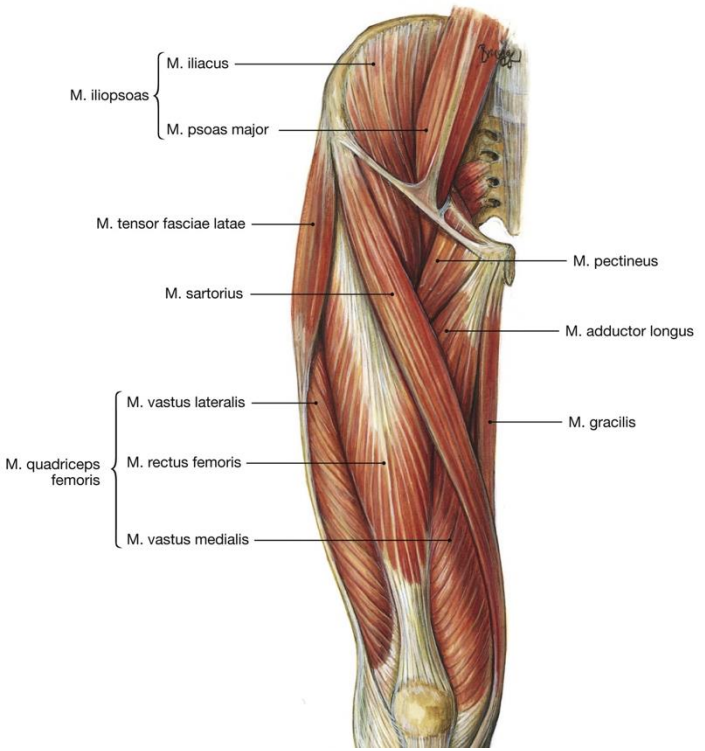
Gluteus medius/minimus på standbenet trækker bækkenet op
Stance-side abductors hold the pelvis level

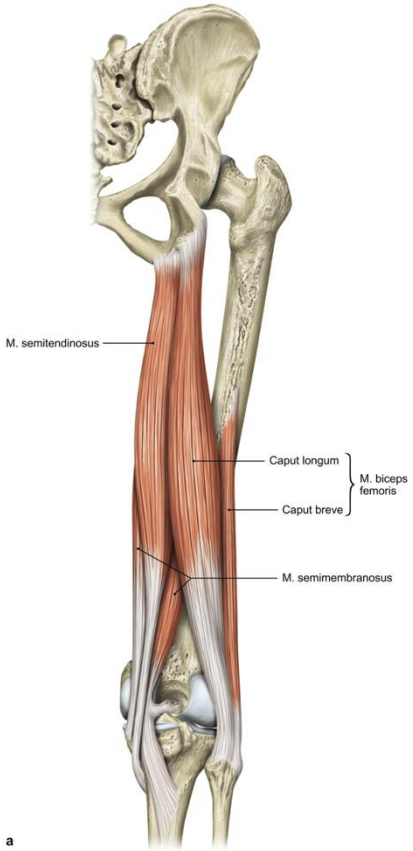
Klinik: Positiv test: bækkenet falder på den side, der IKKE står på. Kompensation: overkroppen læner mod standbenet.
Positive test: the pelvis drops on the side NOT standing. Compensation: trunk leans toward the stance leg.

Lårets muskler

Muscles of the thigh

Loge *Compartment*	**Muskler**	**Funktion**
Forreste *Anterior*	*M. quadriceps femoris* (rectus femoris, vastus medialis, intermedius, lateralis), *m. sartorius*	Knæekstension; hoftefleksion (rectus, sartorius)
Mediale *Medial*	*Mm. adductores* (longus, brevis, magnus), *m. gracilis*, *m. pectineus*	Adduktion af låret
Bageste *Posterior*	*M. biceps femoris*, *m. semitendinosus*, *m. semimembranosus*	Hofteekstension og knæfleksion





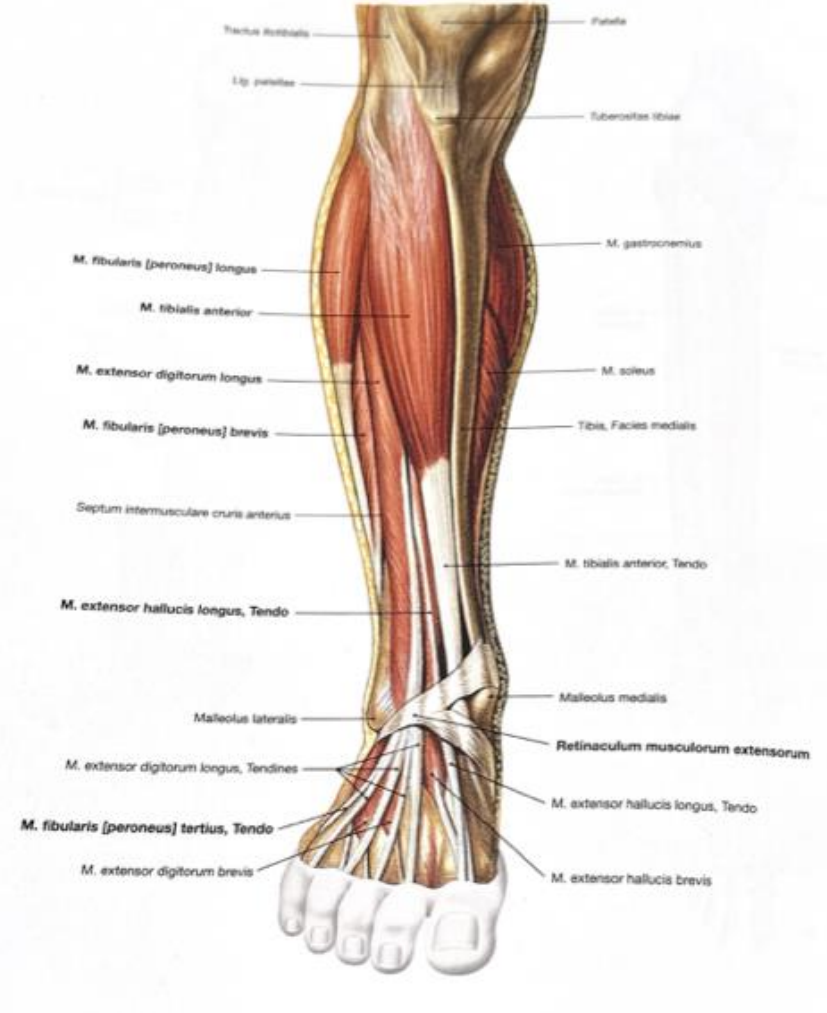
Klinik: Adduktorstræk ('lyskeskade') rammer oftest m. adductor longus ved sparkebevægelser.
Adductor strain ('groin injury') most often affects adductor longus in kicking.

Klinik: Hamstringstræk er hyppigt ved sprint; proksimal avulsion ved tuber ischiadicum er alvorlig.
Hamstring strain is common in sprinting; proximal avulsion at the ischial tuberosity is serious.

Underbenets fire muskelloger

The four compartments of the leg

Loge *Compartment*	**Muskler**	**Funktion**
Forreste *Anterior*	*M. tibialis anterior*, *m. extensor digitorum longus*, *m. extensor hallucis longus*, *m. fibularis tertius*	Dorsalfleksion, tåekstension, inversion
Laterale *Lateral*	*M. fibularis longus* og *brevis*	Eversion
Overfladisk bageste *Superficial posterior*	*M. gastrocnemius*, *m. soleus*, *m. plantaris*	Plantarfleksion
Dybe bageste *Deep posterior*	*M. tibialis posterior*, *m. flexor digitorum longus*, *m. flexor hallucis longus*, *m. popliteus*	Inversion, tåfleksion; popliteus 'låser knæet op'



Ræsonnement: Hver loge har sin egen arterie og sin egen funktion – derfor peger den bevægelse, der svigter, på hvilke muskler der er påvirket.

Each compartment has its own artery and function, so the movement that fails points to the muscles involved.

Triceps surae og akillessenen

Triceps surae and the Achilles tendon

M. gastrocnemius

Gastrocnemius

To hoveder fra femurs kondyler

Two heads from the femoral condyles

Krydser knæ og ankel

Crosses knee and ankle

Kraftigst ved strakt knæ

Strongest with the knee extended

M. soleus

Soleus

Fra tibia og fibula

From tibia and fibula

Krydser kun ankel

Crosses the ankle only

Posturalt og venepumpe

Postural and a venous pump

Tendo calcaneus

Achilles tendon

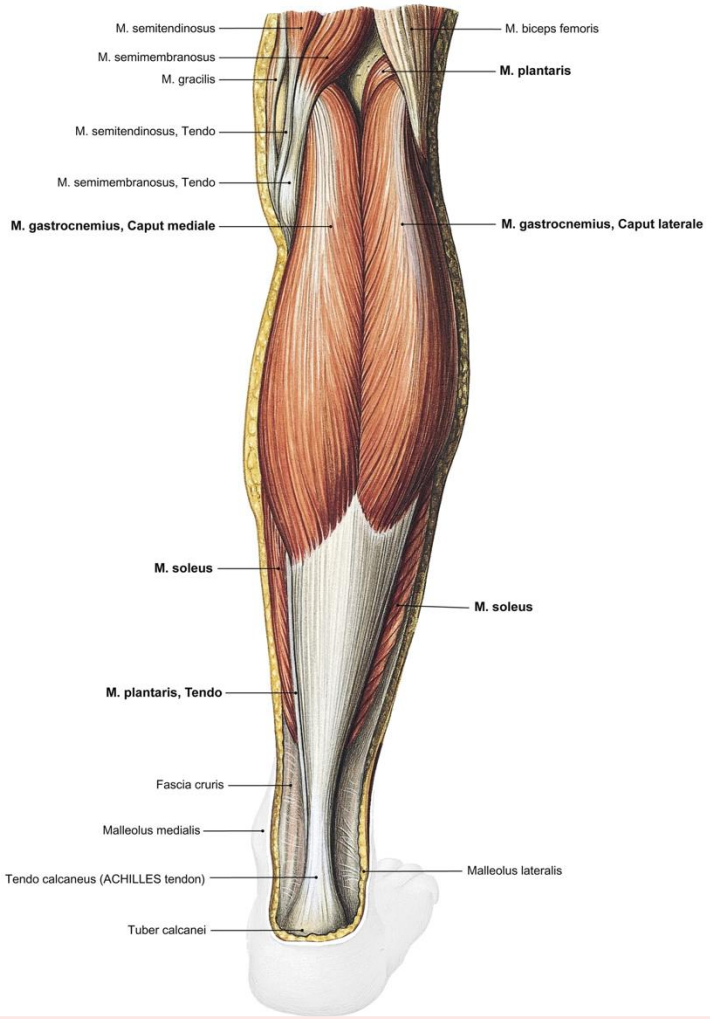
Kroppens kraftigste sene

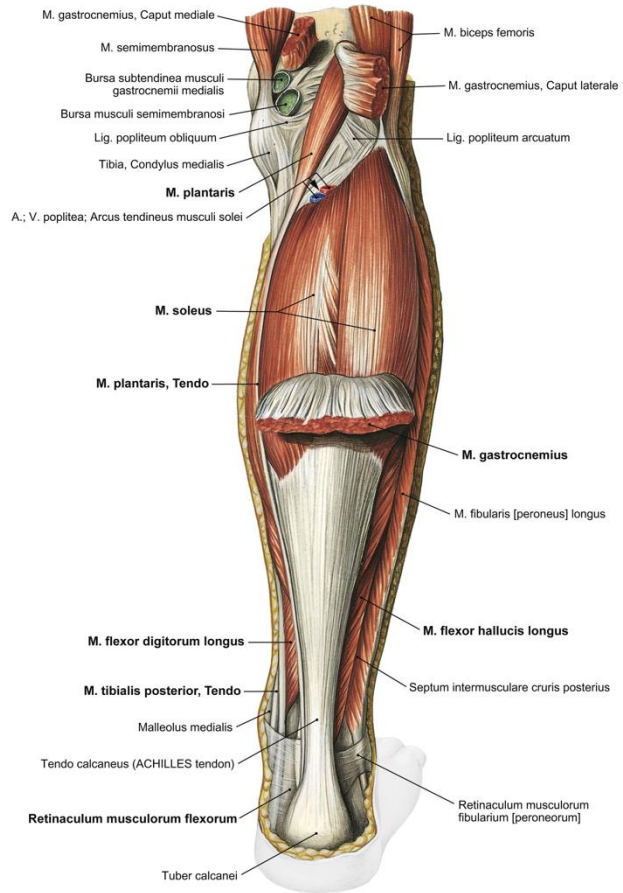
The body's strongest tendon

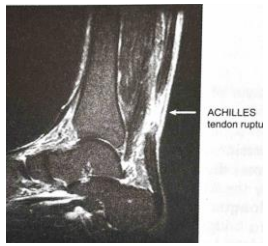
Hæfter på *tuber calcanei*

Inserts on the calcaneal tuberosity

Bursa retrocalcanea







Klinik: Akillesruptur: pludseligt 'knald', manglende plantarfleksion i tåstand, palpabel defekt. Thompsons test er positiv, når læggen ikke løfter foden.

Achilles rupture: sudden 'snap', inability to plantar flex on tiptoe, palpable gap. Thompson's test is positive when the foot does not move on a calf squeeze.

Fodens muskler

Muscles of the foot

Dorsum pedis

Dorsum of the foot

M. extensor digitorum brevis

M. extensor hallucis brevis

Lag *Layer*	Muskler
1.	*M. abductor hallucis*, *m. flexor digitorum brevis*, *m. abductor digiti minimi*
2.	*M. quadratus plantae*, *mm. lumbricales*
3.	*M. flexor hallucis brevis*, *m. adductor hallucis*, *m. flexor digiti minimi brevis*
4.	*Mm. interossei*

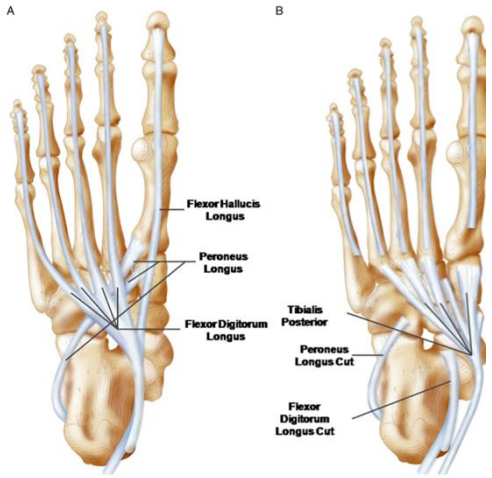


Figure 5 Insertions of the extrinsic foot muscle tendons on the plantar surface of the foot. (A) The insertions of the flexor digitorum longus, flexor hallucis longus and peroneus longus are depicted. Note the longitudinal alignment of the flexor tendons as it relates to their functional contributions to longitudinal foot stability. The oblique alignment of the peroneus longus tendon and its midfoot orientation clearly supports the transverse arch. (B) The insertion of the tibialis posterior tendon is depicted with the tendons of figure 1A cut away. Note the widespread insertions of the tibialis posterior tendon across the tarsals and metatarsals elucidating its functional contributions to longitudinal and transverse arch stability.



Figure 6 The intrinsic foot muscles are presented in their anatomic orientation within the four plantar layers and the dorsal intrinsic muscle. The numbers correspond to the muscles as follows: (1) abductor hallucis, (2) flexor digitorum brevis, (3) abductor digiti minimi, (4) quadratus plantae (note its insertion into the flexor digitorum tendon), (5) lumbricals (note their origin from the flexor digitorum longus tendon), (6) flexor digiti minimi, (7) adductor hallucis oblique (a) and transverse (b) heads, (8) flexor hallucis brevis, (9) plantar interossei, (10) dorsal interossei and (11) extensor digitorum brevis.

Klinik: "Foot core": fodens intrinsiske muskler stabiliserer den mediale længdebue indefra. Korte-fod-øvelsen (metatarsalhovederne trækkes mod hælen uden tåkrølning) bruges til at træne dem.

"Foot core": the intrinsic foot muscles stabilise the medial longitudinal arch from within. The short foot exercise (drawing the metatarsal heads towards the heel without curling the toes) is used to retrain them

McKeon PO, Hertel J, Bramble D, Davis I. The foot core system: a new paradigm for understanding intrinsic foot muscle function. Br J Sports Med. 2015;49(5):290.

Læringsmål

Learning objectives

1. Redegøre for de vigtigste knogler og anatomiske landemærker i underekstremiteten
   Describe the major bones and anatomical landmarks of the lower extremity

2. Redegøre for de vigtigste led og ligamenter (hofte, knæ, ankel og fod) samt forklare deres betydning for bevægelse og stabilitet
   Describe the major joints and ligaments (hip, knee, ankle and foot) and explain their role in movement and stability

3. **Redegøre for de vigtigste muskelgrupper omkring hofte og balle samt forklare deres rolle i bevægelse og stabilisering af bækkenet**
   Describe the major muscle groups of the hip and gluteal region and explain their role in movement and pelvic stabilization

4. **Redegøre for de vigtigste muskelgrupper i lår og underben og relatere deres placering til deres funktion**
   Describe the major muscle groups of the thigh and leg and relate their position to their function

5. **Redegøre for de vigtigste strukturer i foden, herunder fodbuer, plantaraponeurosen og funktionelt vigtige fodmuskler**
   Describe the major structures of the foot, including the arches, plantar aponeurosis and functionally important foot muscles

Litteratur og læsning

Reading

Bevægeapparatets anatomi (modulets danske lærebog)
The Danish textbook for the module

Sobotta Atlas of Anatomy: Underekstremitet
Lower limb volume

Muskelskema (modulets materiale på Moodle)
Muscle table, course material

Modul 5.3 Bevægeapparatet

Underekstremitetens (LE) – regionale anatomi

Lower extremity – regional anatomy

Sådan læser du slides

Normal tekst = dansk

Kursiv tekst = latinsk anatomisk term

Kursiv, mindre og grøn = engelsk

Tekstbøger: Bevægeapparatets anatomi og Sobotta Atlas of Anatomy

Department of Health Science and Technology, Aalborg University

Fra systematisk til regional anatomi

From systematic to regional anatomy

Systematisk anatomi: Hvad er strukturerne?
Systematic anatomy: What are the structures?

Regional anatomi: Hvor ligger strukturerne i forhold til hinanden?
Regional anatomy: Where are structures located relative to one another?

Muskler bruges som grænser og landemærker
Muscles are used as boundaries and landmarks

Kar og nerver følges gennem regioner
Vessels and nerves are traced through regions

Passager (kanaler, huller, tunneler) er klinisk vigtige
Passageways (canals, foramina, tunnels) are clinically important

Læringsmål

Learning objectives

1. **Redegøre for afgrænsninger og indhold i regio glutealis, herunder foramen supra- og infrapiriforme og forløbet af n. ischiadicus**
   Describe the boundaries and contents of the gluteal region, including the supra- and infrapiriform foramina and the course of the sciatic nerve

2. **Redegøre for afgrænsninger, gulv, loft og indhold i trigonum femorale og canalis adductorius, herunder lacuna musculorum, lacuna vasorum og canalis femoralis**
   Describe the boundaries, floor, roof and contents of the femoral triangle and adductor canal, including the lacunae and the femoral canal

3. **Redegøre for afgrænsninger og indhold i fossa poplitea, herunder forholdet mellem a. poplitea, v. poplitea og n. tibialis**
   Describe the boundaries and contents of the popliteal fossa, including the relation of the popliteal artery, vein and tibial nerve

4. **Redegøre for de vigtigste regionale relationer i regio cruris, herunder de fire muskelloger samt forløbet af kar og perifere nerver**
   Describe the major regional relationships in the leg, including the four compartments and the course of vessels and peripheral nerves

5. **Redegøre for underekstremitetens arterielle forsyning, venøse og lymfatiske drænage samt kutane innervation (dermatomer)**
   Describe the arterial supply, venous and lymphatic drainage and cutaneous innervation (dermatomes) of the lower limb

6. **Redegøre for de vigtigste passager i regio tarsalis og pedis, herunder tarsaltunnelen, retinacula og planta pedis**
   Describe the major passageways of the ankle and foot, including the tarsal tunnel, retinacula and the sole of the foot

Regioner i underekstremiteten

Regions of the lower limb

Balleregionen
Gluteal region

Regio glutealis

N. ischiadicus, foramen supra- og infrapiriforme
Sciatic nerve, piriform foramina

Klinik: injektioner, nerveskade
Clinic: injections, nerve injury

Låret
Thigh

Regio femoris

Trigonum femorale, canalis adductorius
Femoral triangle, adductor canal

Klinik: arteriepuls, hernier
Clinic: arterial pulse, hernias

Knæet
Knee

Regio genus

Fossa poplitea, pes anserinus
Popliteal fossa, pes anserinus

Klinik: aneurismer, Bakers cyste
Clinic: aneurysm, Baker's cyst

Underbenet
Leg

Regio cruris

Fire muskelloger, n. fibularis communis
Four compartments, common fibular nerve

Klinik: logesyndrom, dropfod
Clinic: compartment syndrome, foot drop

Ankel og fod
Ankle and foot

Regio tarsalis et pedis

Tarsaltunnel, retinacula, planta pedis
Tarsal tunnel, retinacula, sole

Klinik: tunnelsyndrom, tryksår
Clinic: tunnel syndrome, ulcers

Balleregionen

Regio glutealis · Gluteal region

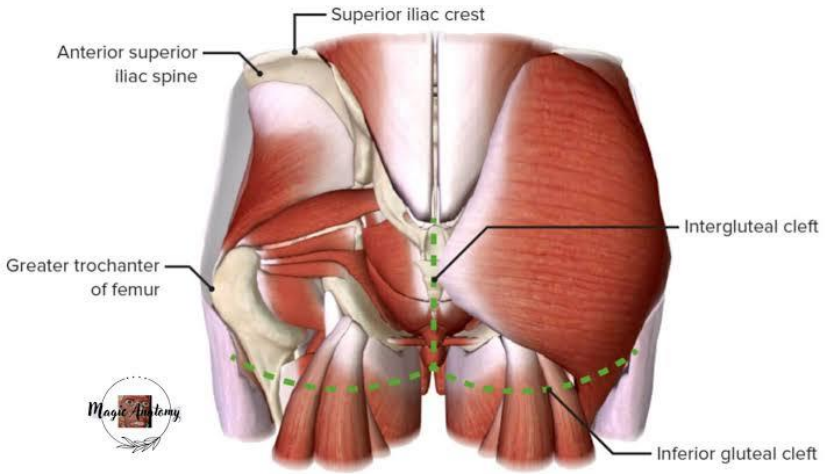
Afgrænsning
Boundaries

Oven: *crista iliaca*
Superior: iliac crest

Nedenfor: *sulcus glutealis*
Inferior: gluteal fold

Medialt: *sacrum* og *coccyx*
Medial

Lateralt: *trochanter major*
Lateral



Lag
Layers

Overfladisk: *m. gluteus maximus*
Superficial

Dybt: *m. gluteus medius* og *minimus*
Deep

Dybest: *m. piriformis* og laterale rotatorer
Deepest: piriformis and lateral rotators

Nøgleligamenter
Key ligaments

Lig. sacrotuberale

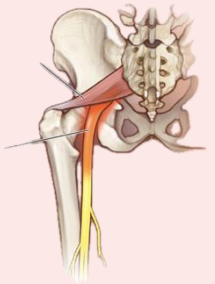
Lig. sacrospinale

Danner *foramen ischiadicum majus* og *minus*
Form the greater and lesser sciatic foramina

Klinik: Piriformis-syndrom: smerter i balden med udstråling langs n. ischiadicus ved irritation/kompression i den dybe glutealregion. Anatomiske varianter af m. piriformis/n. ischiadicus ses hos ca. 17 %, men lige så ofte hos raske som hos patienter, så varianter forklarer ikke syndromet alene.

Piriformis syndrome: gluteal pain radiating along the sciatic nerve from irritation/compression in the deep gluteal region. Variants of piriformis/sciatic nerve anatomy occur in ~17%, but equally often in healthy cadavers and in patients, so variants alone do not explain the syndrome.

Smoll, Clin Anat 2010;23:8 · Lopez Castellanos & Feipel, Morphologie 2026;110:101114



Foramen supra- og infrapiriforme

Suprapiriform and infrapiriform foramina

Foramen suprapiriforme

Over m. piriformis

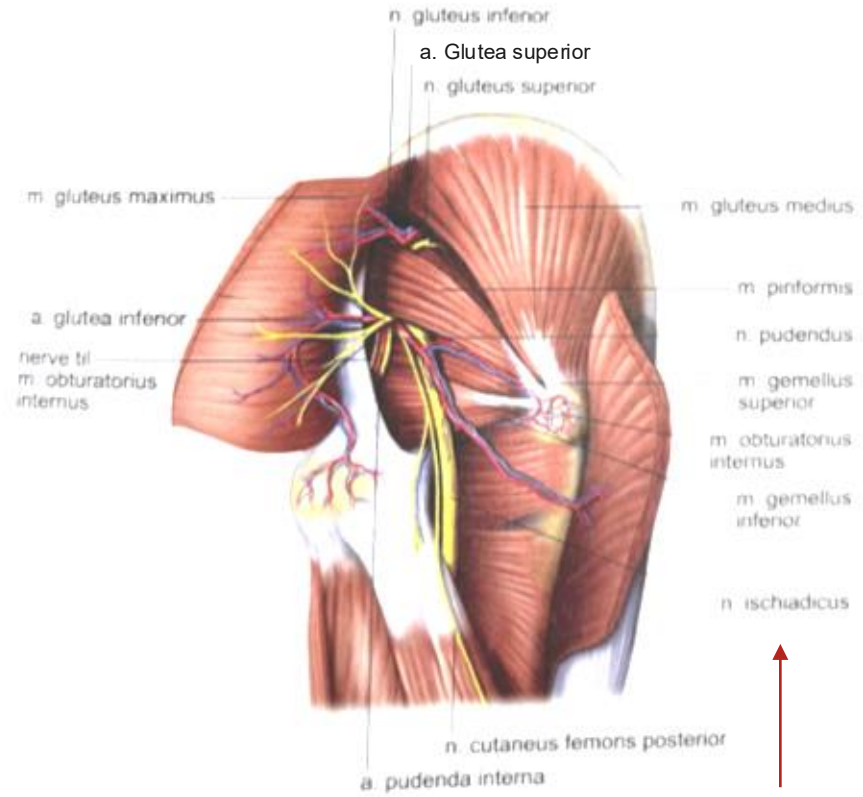
N. gluteus superior
A. og v. glutea superior

M. piriformis

Foramen infrapiriforme

Under m. piriformis

N. ischiadicus, n. cutaneus femoris posterior
N. gluteus inferior, a. og v. glutea inferior
N. pudendus, a. og v. pudenda interna



Figur 19. Kar og nerver i spatium subgluteale.

Hvorfor er det vigtigt?

Why does it matter?

Forholdet til piriformis er en orienteringsregel

The relation to piriformis is an orientation rule

Nervernes placering afgør, hvor man kan injicere

Nerve position determines safe injection sites

Klinik: Piriformis-syndrom: kompression af n. ischiadicus kan give balle- og benssmerter. Skal skelnes fra lumbal radikulopati.

Piriformis syndrome: compression of the sciatic nerve may give buttock and leg pain. Must be distinguished from lumbar radiculopathy.

Læringsmål

Learning objectives

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   Describe the boundaries and contents of the gluteal region, including the supra- and infrapiriform foramina and the course of the sciatic nerve

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   Describe the boundaries, floor, roof and contents of the femoral triangle

3. Redegøre for afgrænsninger og indhold i fossa poplitea, herunder forholdet mellem a. poplitea, v. poplitea og n. tibialis
   Describe the boundaries and contents of the popliteal fossa, including the relation of the popliteal artery, vein and tibial nerve

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   Describe the major regional relationships in the leg, including the four compartments and the course of vessels and peripheral nerves

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   Describe the major passageways of the ankle and foot, including the tarsal tunnel, retinacula and the sole of the foot

Trigonum femorale

The femoral triangle

Base: *lig. inguinale*
Base: inguinal ligament

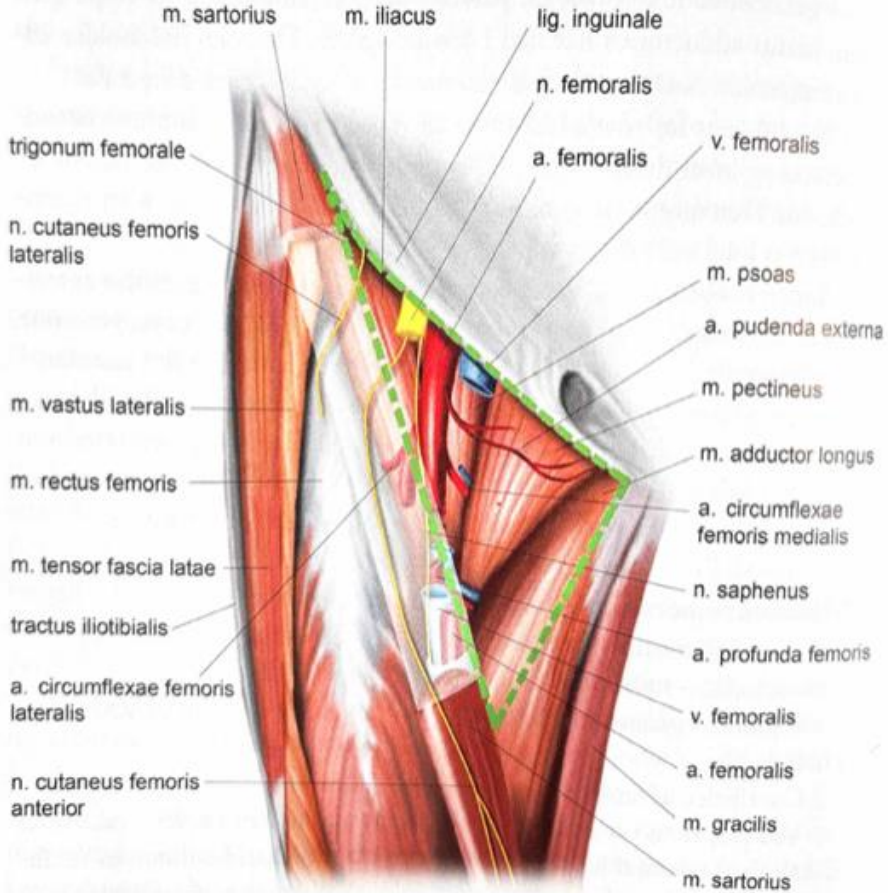
Lateral: *m. sartorius*; medial: *m. adductor longus*
Lateral and medial borders

Gulv: *m. iliopsoas* og *m. pectineus*
Floor

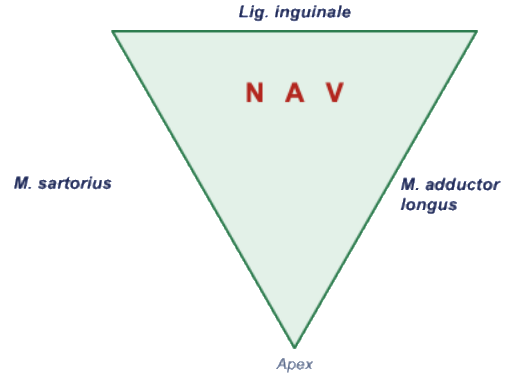
Tag: *fascia lata* med *hiatus saphenus*
Roof

Indhold lateral→medial: **N**ervus, **A**rterie, **V**ene, **E**mpty space (lymfe)
Contents lateral to medial: nerve, artery, vein, empty space

https://teachmeanatomy.info/lower-limb/areas/femoral-triangle/



Figur 21. Trigonum femorale.



Klinik: A. femoralis kan palperes ved midtpunktet mellem SIAS og symphysis pubica. Bruges til arteriepunktur, kateter og femoralisblokade.
The femoral artery is palpable at the midpoint between the ASIS and pubic symphysis. Used for arterial puncture, catheters and femoral nerve block.

https://www.ncbi.nlm.nih.gov/books/NBK541140/

Læringsmål

Learning objectives

1. Redegøre for afgrænsninger og indhold i regio glutealis, herunder foramen supra- og infrapiriforme og forløbet af n. ischiadicus
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Fossa poplitea

The popliteal fossa

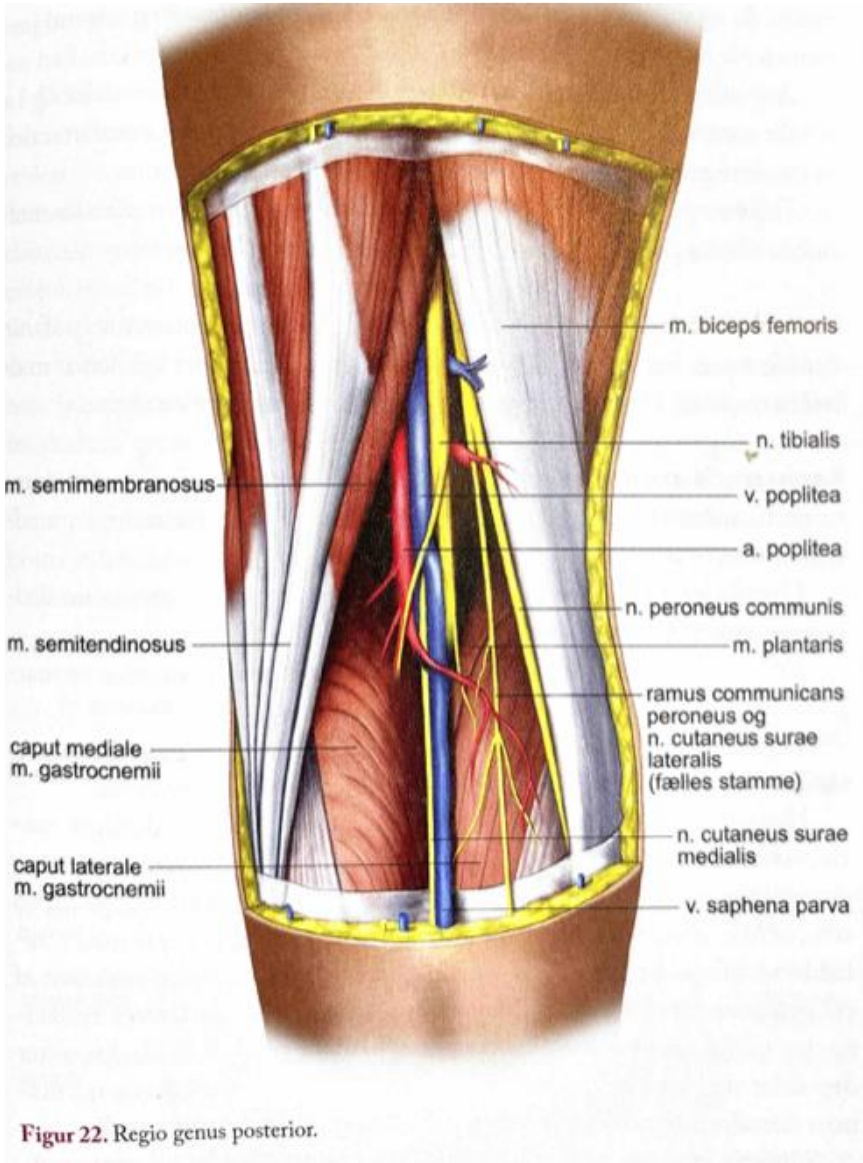
Superolateralt: *m. biceps femoris*
Superolateral

Superomedialt: *m. semimembranosus* og *semitendinosus*
Superomedial

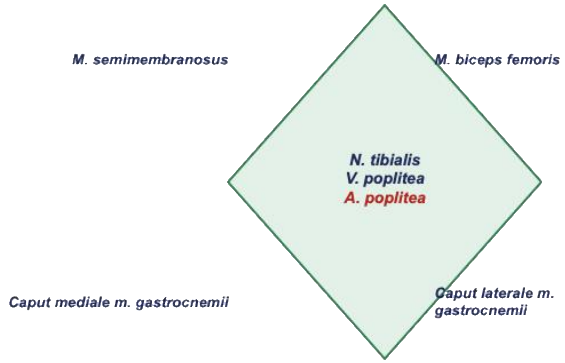
Inferiort: *caput mediale* og *laterale m. gastrocnemii*
Inferior

Gulv: *planum popliteum*, *lig. popliteum obliquum*, *m. popliteus*
Floor

Tag: *fascia poplitea*
Roof



Figur 22. Regio genus posterior.



Fossa poplitea: indhold og klinik

Popliteal fossa: contents and clinical relevance

Fra overfladisk til dyb

Superficial to deep

N. tibialis (mest overfladisk)
Most superficial

V. poplitea

A. poplitea (dybest, nærmest femur)
Deepest, next to the femur

Andet indhold

Other contents

N. fibularis communis langs biceps' mediale sene
Along the medial border of the biceps tendon

V. saphena parva gennemborer *fascia poplitea*
The small saphenous vein pierces the roof

Lymfeknuder
Popliteal lymph nodes

Klinik: Popliteaaneurisme er den hyppigste perifere aneurismelokalisation. Kan trombosere.
Popliteal aneurysm is the most common peripheral aneurysm and can thrombose.

Klinik: Knæluksation kan skade a. poplitea og kræver vaskulær vurdering.
Knee dislocation can injure the popliteal artery and requires vascular assessment.

Klinik: Bakers cyste: udposning bag knæet, ofte ved ledeffusion.
Baker's cyst: swelling behind the knee, often with joint effusion.

Popliteal Artery Aneurysm Visualized on Magnetic Resonance Angiography. Magnetic resonance angiography visualizes a popliteal artery aneurysm. Contributed by S Bhimji, MD



Thrombosed Popliteal Aneurysm
Kaohinani J. Longwolf; Jeffery B. Dattilo.
https://www.ncbi.nlm.nih.gov/books/NBK541115/

Knæregionens landemærker

Landmarks of the knee region

Pes anserinus

Goose's foot

M. sartorius, *m. gracilis*, *m. semitendinosus*

Hæfter anteromedialt på tibia

Insert anteromedially on the tibia

Bursa anserina under senerne

Bursa beneath the tendons

Lateralt

Lateral

Tractus iliotibialis til tibia

Iliotibial tract to the tibia

Caput fibulae og *n. fibularis communis*

M. biceps femoris-sene

Biceps tendon

Fortil

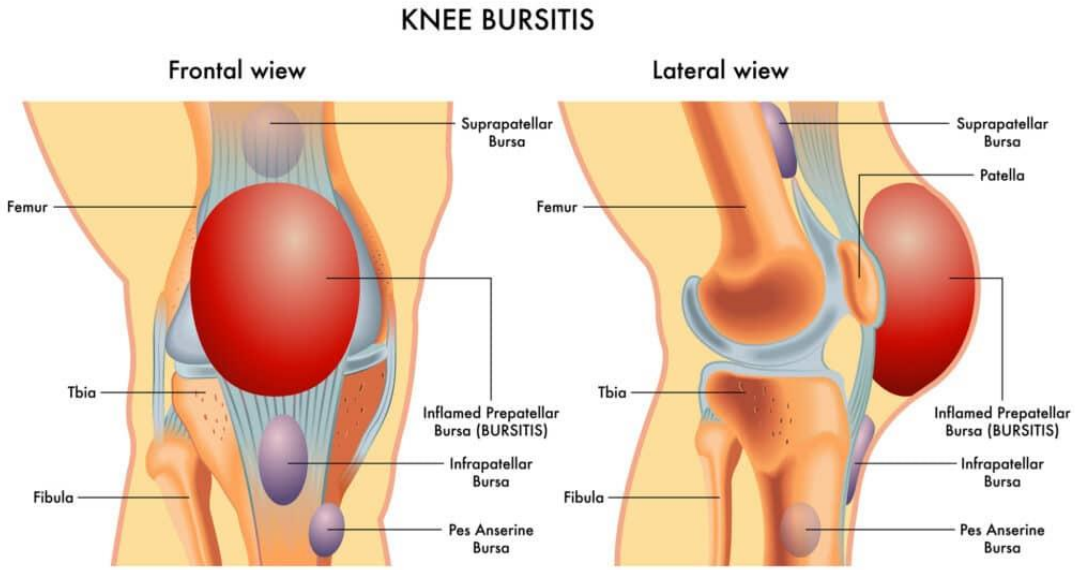
Anterior

Lig. patellae til *tuberositas tibiae*

Bursa prepatellaris

Patellarefleks: L3–L4

Patellar reflex



Klinik: Pes anserin-bursitis giver smerte medialt under ledlinjen. Prepatellar bursitis opstår ved knælende arbejde.

Pes anserine bursitis causes pain medially below the joint line. Prepatellar bursitis follows kneeling work.

Læringsmål

Learning objectives

1. Redegøre for afgrænsninger og indhold i regio glutealis, herunder foramen supra- og infrapiriforme og forløbet af n. ischiadicus
   Describe the boundaries and contents of the gluteal region, including the supra- and infrapiriform foramina and the course of the sciatic nerve

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4. Redegøre for de vigtigste regionale relationer i regio cruris, herunder de fire muskelloger samt forløbet af kar og perifere nerver
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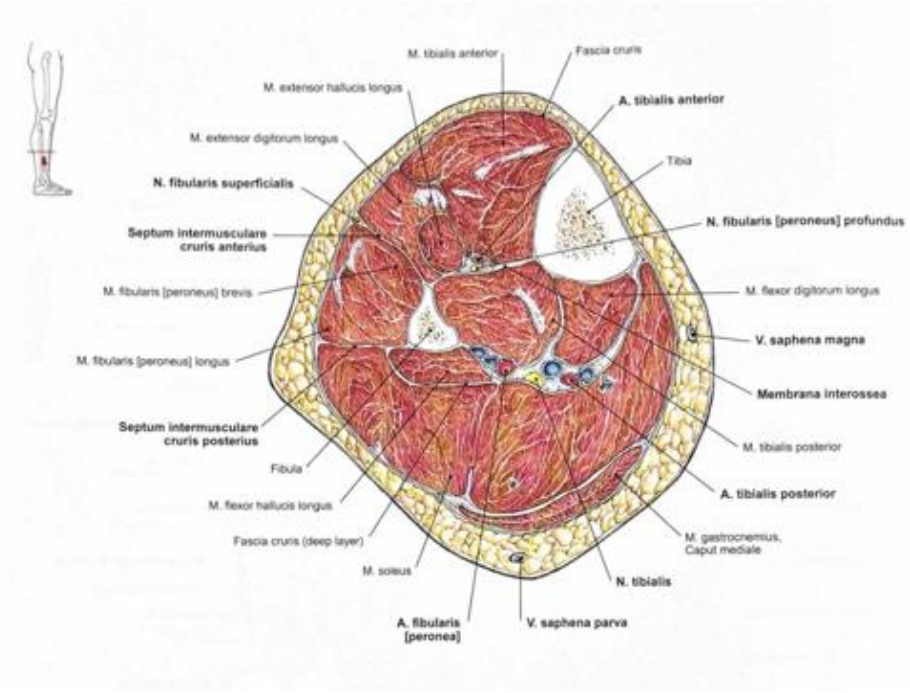
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   Describe the major passageways of the ankle and foot, including the tarsal tunnel, retinacula and the sole of the foot

Regio cruris: de fire muskelloger

The leg: four fascial compartments

Crural fascia, *membrana interossea* og septa adskiller logerne
The fascia, interosseous membrane and septa separate the compartments

Hver loge har egne muskler, egen nerve og egen arterie
Each compartment has its own muscles, nerve and artery



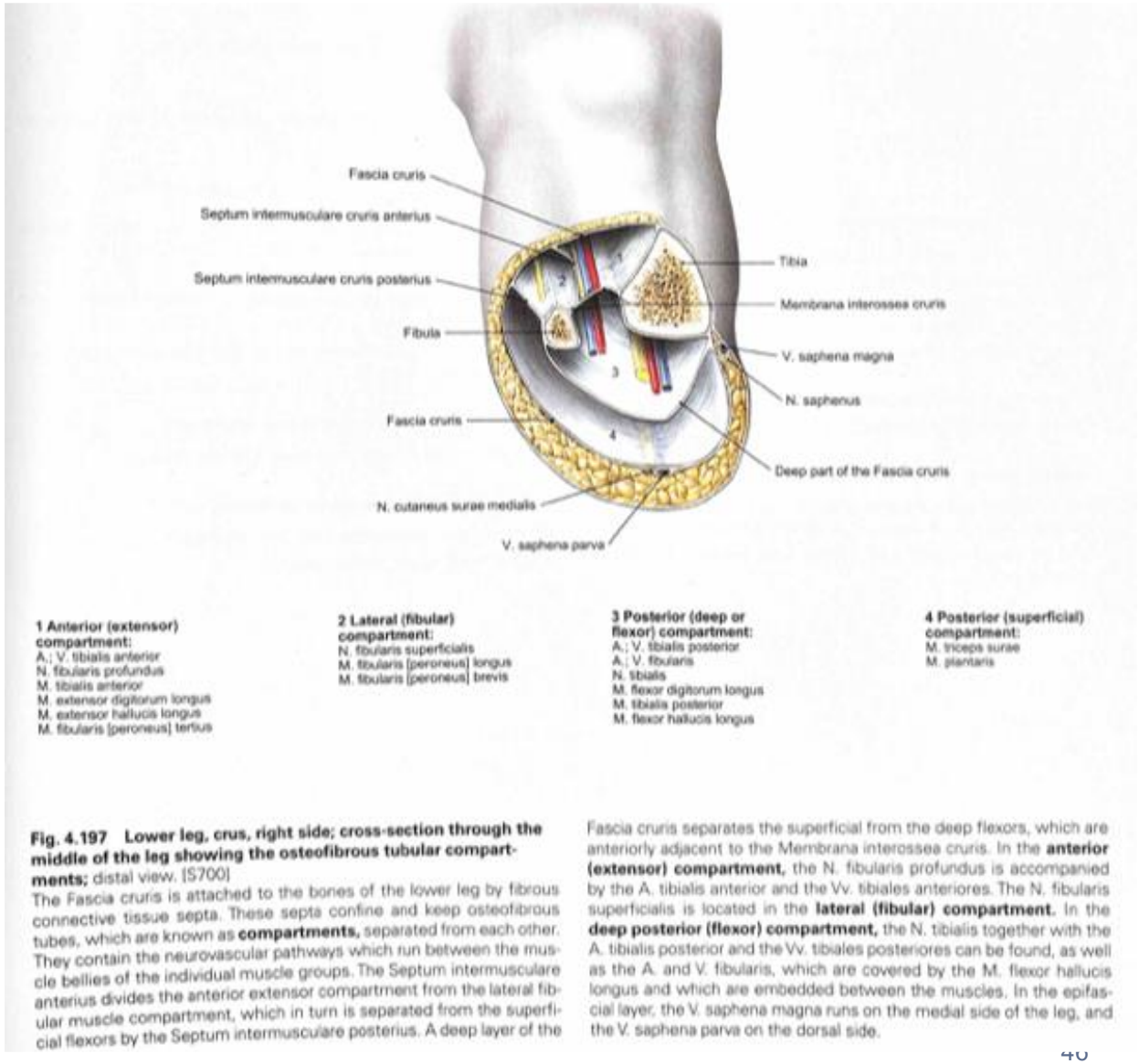


Fig. 4.197 Lower leg, crus, right side; cross-section through the middle of the leg showing the osteofibrous tubular compartments; distal view. [S700]
The Fascia cruris is attached to the bones of the lower leg by fibrous connective tissue septa. These septa confine and keep osteofibrous tubes, which are known as **compartments,** separated from each other. They contain the neurovascular pathways which run between the muscle bellies of the individual muscle groups. The Septum intermusculare anterius divides the anterior extensor compartment from the lateral fibular muscle compartment, which in turn is separated from the superficial flexors by the Septum intermusculare posterius. A deep layer of the Fascia cruris separates the superficial from the deep flexors, which are anteriorly adjacent to the Membrana interossea cruris. In the **anterior (extensor) compartment,** the N. fibularis profundus is accompanied by the A. tibialis anterior and the Vv. tibiales anteriores. The N. fibularis superficialis is located in the **lateral (fibular) compartment.** In the **deep posterior (flexor) compartment,** the N. tibialis together with the A. tibialis posterior and the Vv. tibiales posteriores can be found, as well as the A. and V. fibularis, which are covered by the M. flexor hallucis longus and which are embedded between the muscles. In the epifascial layer, the V. saphena magna runs on the medial side of the leg, and the V. saphena parva on the dorsal side.

Logernes indhold, nerver og kar

Contents, nerves and vessels of the compartments

Loge *Compartment*	**Muskler**	**Nerve og arterie**	**Funktion**
Forreste *Anterior*	*M. tibialis anterior*, *m. extensor hallucis longus*, *m. extensor digitorum longus*, *m. fibularis tertius*	*N. fibularis profundus*; *a. tibialis anterior*	Dorsalfleksion, tåekstension
Laterale *Lateral*	*M. fibularis longus* og *brevis*	*N. fibularis superficialis*	Eversion
Overfladisk bageste *Superficial posterior*	*M. gastrocnemius*, *m. soleus*, *m. plantaris*		Plantarfleksion
Dybe bageste *Deep posterior*	*M. tibialis posterior*, *m. flexor digitorum longus*, *m. flexor hallucis longus*	*N. tibialis*; *a. tibialis posterior* og *a. fibularis*	Inversion, tåfleksion



Compression syndrome (compartment syndrome) most often occurs in the anterior (extensor) compartment, less often in the lateral (fibular) compartment.

a

b

a Post-traumatic swelling of the extensor muscles or a long walk can compress and damage the supplying vessels and nerves to the muscles. In addition to pain, there may also be a pulse deficit of the A. dorsalis pedis, originating in the A. tibialis anterior. But most frequently, the compression causes a lesion of the N. fibularis profundis (Clinical remarks → Fig. 4.154) with resulting functional (extension) deficits in the talocrural joint and loss of sensory perception in the first interdigital space. [E993]

b The treatment requires an urgent splitting of the fascia, which, however, requires long-term immobilisation of the open leg. To confirm the diagnosis, the pressure in the compartments should be measured with a pressure sensor. [E475]

Logesyndrom

Compartment syndrome

Mekanisme

Mechanism

Stigende tryk i en lukket loge

Rising pressure in a closed compartment

Nedsat perfusion og iskæmi

Reduced perfusion and ischaemia

Årsager

Causes

Tibiafraktur

Tibial fracture

Knusningsskade, reperfusion

Crush injury, reperfusion

For stram gips

Tight cast

Tegn

Signs

Smerte ud af proportion

Pain out of proportion

Smerte ved passiv strækning

Pain on passive stretch

Paræstesi, spændt hævelse

Paraesthesia, tense swelling

Akut: Perifere pulser er ofte bevarede. Hurtig fasciotomi er nødvendig for at redde muskel og nerve.

Peripheral pulses are often preserved. Prompt fasciotomy is needed to save muscle and nerve.

N. fibularis communis og dropfod

The common fibular nerve and foot drop

N. ischiadicus → **N. fibularis communis** *Collum fibulae* → **N. fibularis profundus og superficialis**

Udfald
Deficits

Dropfod, steppage-gang
Foot drop, steppage gait

Tab af dorsalfleksion og eversion
Loss of dorsiflexion and eversion

Følenedsættelse lateralt på underbenet og fodryggen
Sensory loss over the lateral leg and dorsum of the foot

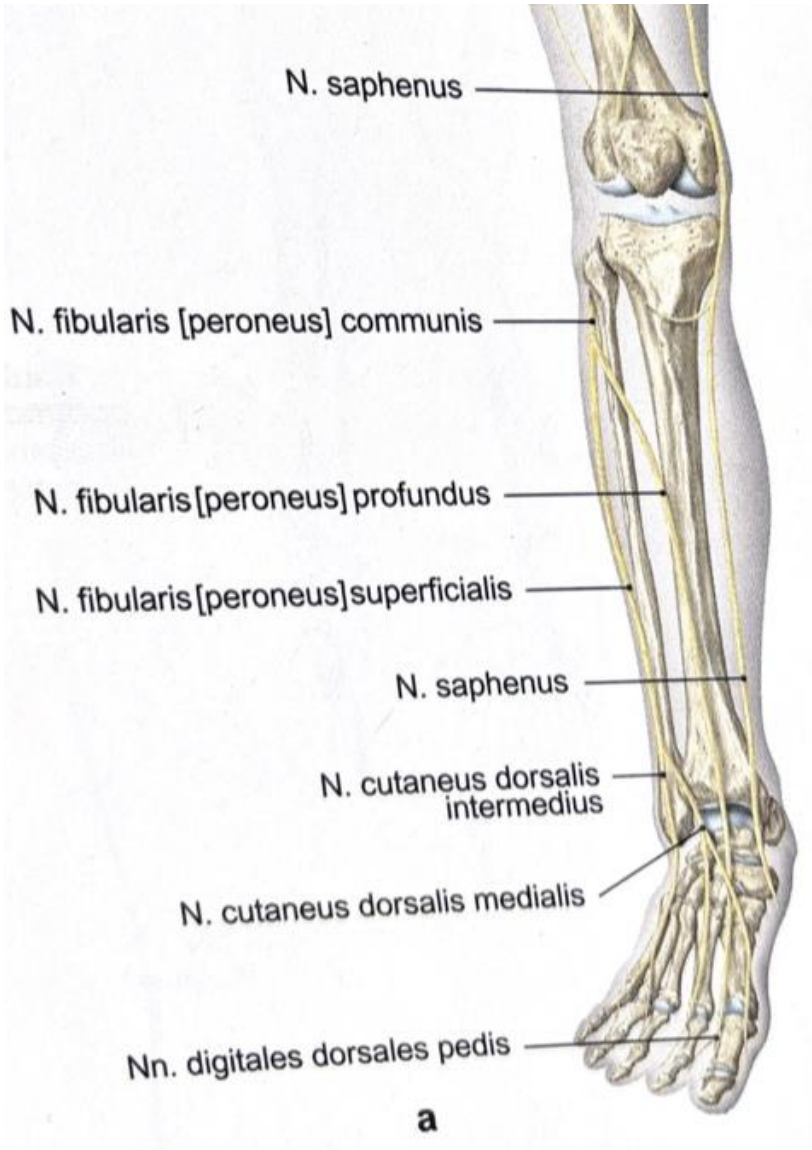
Årsager
Causes

Fraktur af collum fibulae
Fibular neck fracture

Tryk (gips, krydsede ben)
Pressure (cast, crossed legs)

Knæluksation
Knee dislocation

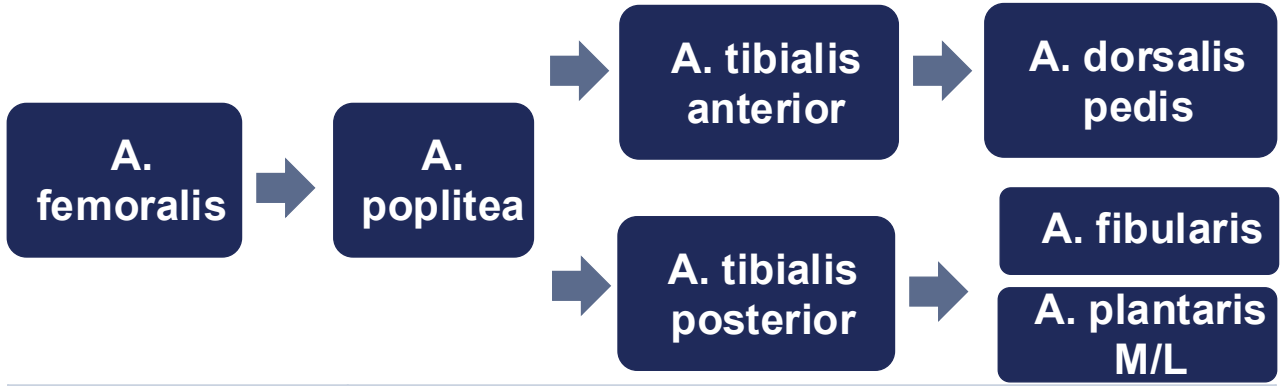
Differentialdiagnose: Ved L5-radikulopati er også inversion (m. tibialis posterior) og hofteabduktion (m. gluteus medius) svækket; ved fibularisparese ved caput fibulae er de bevarede.
In L5 radiculopathy inversion (tibialis posterior) and hip abduction (gluteus medius) are also weak; in fibular palsy at the fibular head they are preserved.



Saber B, Agrawal DK. Foot drop: an anatomical, clinical, and electrodiagnostic approach to localization. J Spine Res Surg. 2026;8(3):50-5.

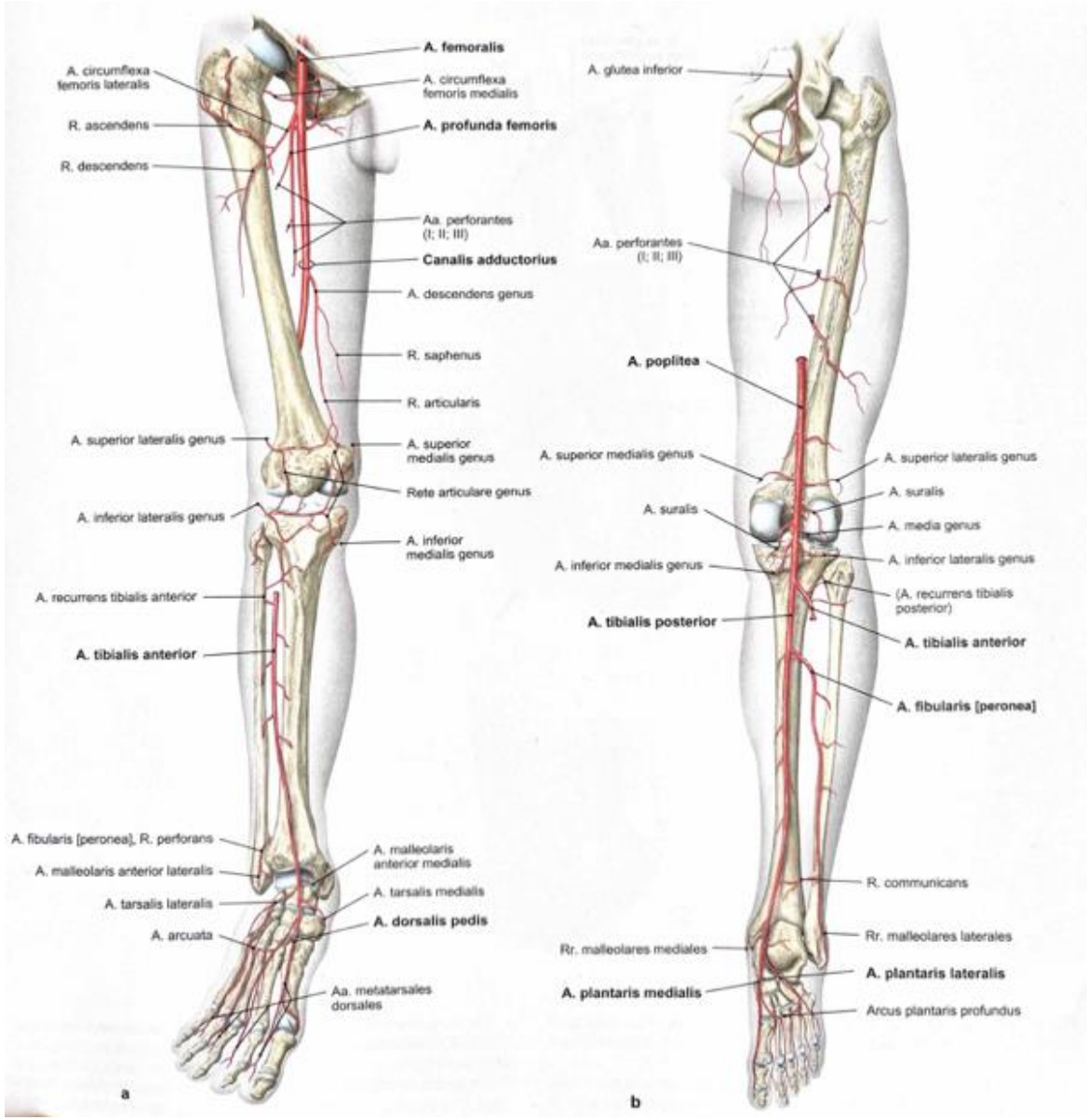
Underekstremitetens arterier

Arteries of the lower limb



Puls *Pulse*	**Palpationssted**
A. femoralis	Midtpunktet mellem SIAS og symphysis pubica
A. poplitea	Dybt i fossa poplitea med let bøjet knæ
A. tibialis posterior	Bag malleolus medialis
A. dorsalis pedis	Lateralt for senen af m. extensor hallucis longus

Klinik: Perifer arteriel sygdom: smerter i læggen ved gang, der lindres i hvile (claudicatio).
Peripheral arterial disease: calf pain on walking relieved by rest (claudication).



Vener og lymfe

Veins and lymphatics

V. saphena magna

Great saphenous vein

Foran malleolus medialis

In front of the medial malleolus

Medialt på ben og lår

Medial leg and thigh

Gennem *hiatus saphenus* til *v. femoralis*

V. saphena parva

Small saphenous vein

Bag malleolus lateralis

Behind the lateral malleolus

Gennem fossa poplitea til *v. poplitea*

Følger *n. suralis*

Accompanies the sural nerve

Lymfe

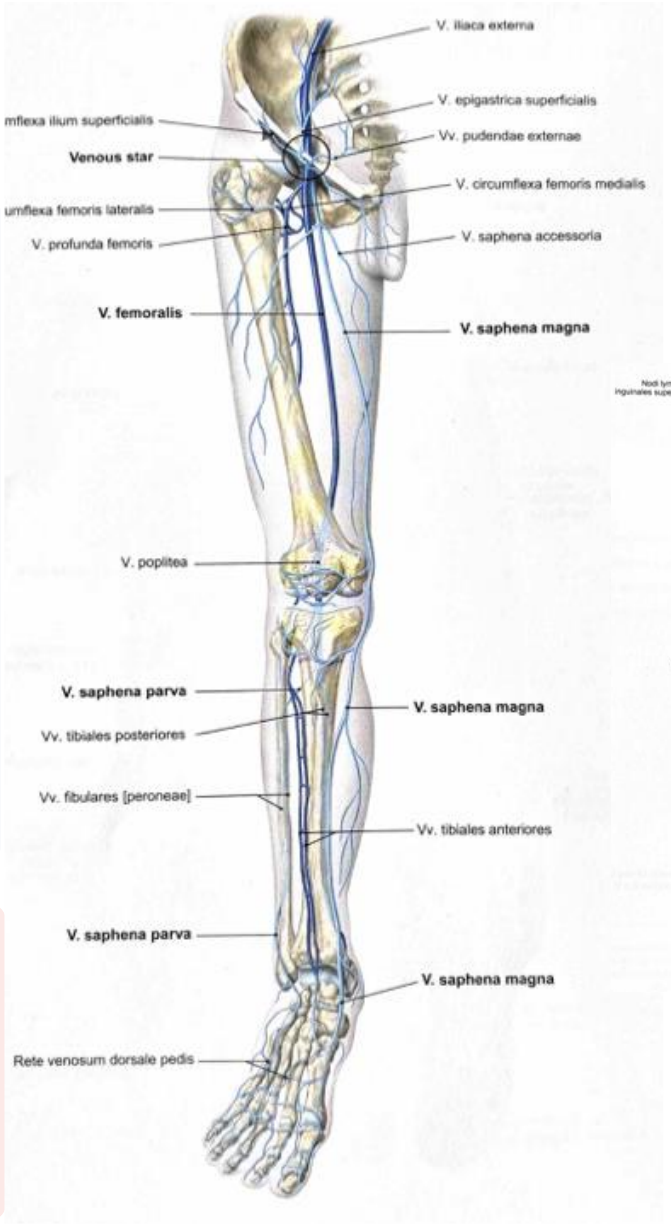
Lymph

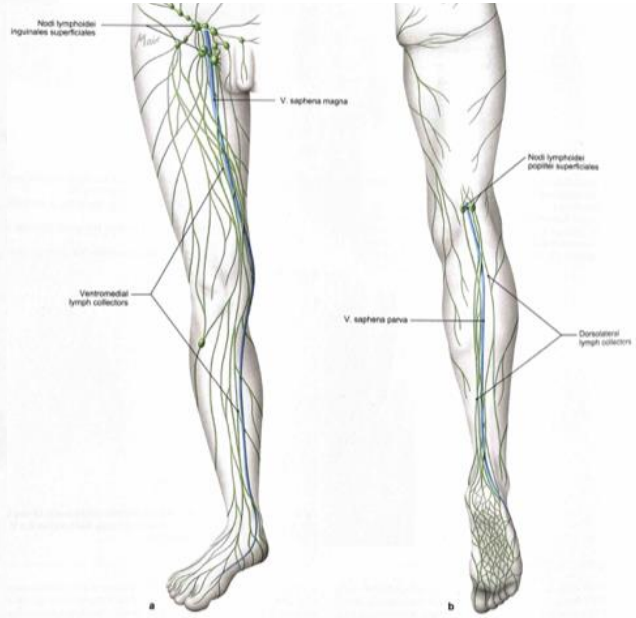
Nodi lymphoidei inguinales

Nodi lymphoidei poplitei

Videre til *nodi iliaci externi*

Then external iliac nodes





Klinik: Åreknuder: klapinsufficiens i saphenavener. DVT opstår typisk i læg- og lårvener.

Varicose veins: valve incompetence in the saphenous veins. DVT typically arises in calf and thigh veins.

Klinik: Svulster og infektioner i fod og ben kan give forstørrede lyskekirtler.

Tumours and infections of the foot and leg can enlarge inguinal nodes.

Dermatomer, myotomer og reflekser

Dermatomes, myotomes and reflexes

dermatomes overlap and vary, and published maps differ (Lee et al., Clin Anat 2008, doi:10.1002/ca.20636; Downs & Laporte, J Orthop Sports Phys Ther 2011, doi:10.2519/jospt.2011.3506). A short footnote on the slide would be consistent with those sources.

Rod *Root*	Dermatom *Dermatome*	Myotom *Movement*	Refleks *Reflex*
L2	Fortil på øvre lår	Hofteflexion; hofteadduktion (L2–L4) [1]	–
L3–L4	Fortil på knæ og mediale underben	Knæekstension; hofteadduktion [1]	Patellarefleks (L3–L4) [2]; især knyttet til L4 [1]
L4	Mediale underben og malleolus medialis	Dorsalfleksion (m. tibialis anterior) (L4–L5)	–
L5	Laterale underben, fodrygg, stortå [1]	Stortåsekstension, tåekstension, dorsalfleksion, inversion og eversion [1]; hofteabduktion [4]	– (asymmetri af medial hamstring-refleks kan tale for L5 [1])
S1	Laterale fod, hæl og fodsål	Plantarfleksion (S1–S2) [1,3]; evt. eversion (L5–S1), hofteekstension, knæfleksion	Akillesrefleks (S1; S1–S2) [2,3]

Sources

1. Dydyk AM, Khan MZ, Singh P. Radicular back pain. StatPearls, 2022. Source of the L2–L4 grouping (including hip adduction), the L5 movements and dermatome, and the S1 plantarflexion.
2. Morimoto T, et al. The usefulness of deep tendon reflexes in the diagnosis of lumbar spine diseases: a narrative review. Cureus. 2024;16(3):e55772. doi:10.7759/cureus.55772
3. Figliuzzi A, Alvarez R, Al-Dhahir MA. Achilles reflex. StatPearls, 2023.
4. Saber B, Agrawal DK. Foot drop: an anatomical, clinical, and electrodiagnostic approach to localization. J Spine Res Surg. 2026;8(3):50-5. doi:10.26502/fjsrs00101 Source for hip abduction and inversion as the key L5 discriminators.
5. Lee MWL, McPhee RW, Stringer MD. An evidence-based approach to human dermatomes. Clin Anat. 2008;21(5):363-73. doi:10.1002/ca.20636
6. Downs MB, Laporte C. Conflicting dermatome maps: educational and clinical implications. J Orthop Sports Phys Ther. 2011;41(6):427-34. doi:10.2519/jospt.2011.3506

Klinik: Mønstret af svaghed, følenedsættelse og reflekser skelner mellem rodskade (L5) og perifer nerveskade (n. fibularis communis).

The pattern of weakness, sensory loss and reflexes separates root lesions from peripheral nerve lesions.

Læringsmål

Learning objectives

1. Redegøre for afgrænsninger og indhold i regio glutealis, herunder foramen supra- og infrapiriforme og forløbet af n. ischiadicus
   Describe the boundaries and contents of the gluteal region, including the supra- and infrapiriform foramina and the course of the sciatic nerve

2. Redegøre for afgrænsninger, gulv, loft og indhold i trigonum femorale
   Describe the boundaries, floor, roof and contents of the femoral triangle

3. Redegøre for afgrænsninger og indhold i fossa poplitea, herunder forholdet mellem a. poplitea, v. poplitea og n. tibialis
   Describe the boundaries and contents of the popliteal fossa, including the relation of the popliteal artery, vein and tibial nerve

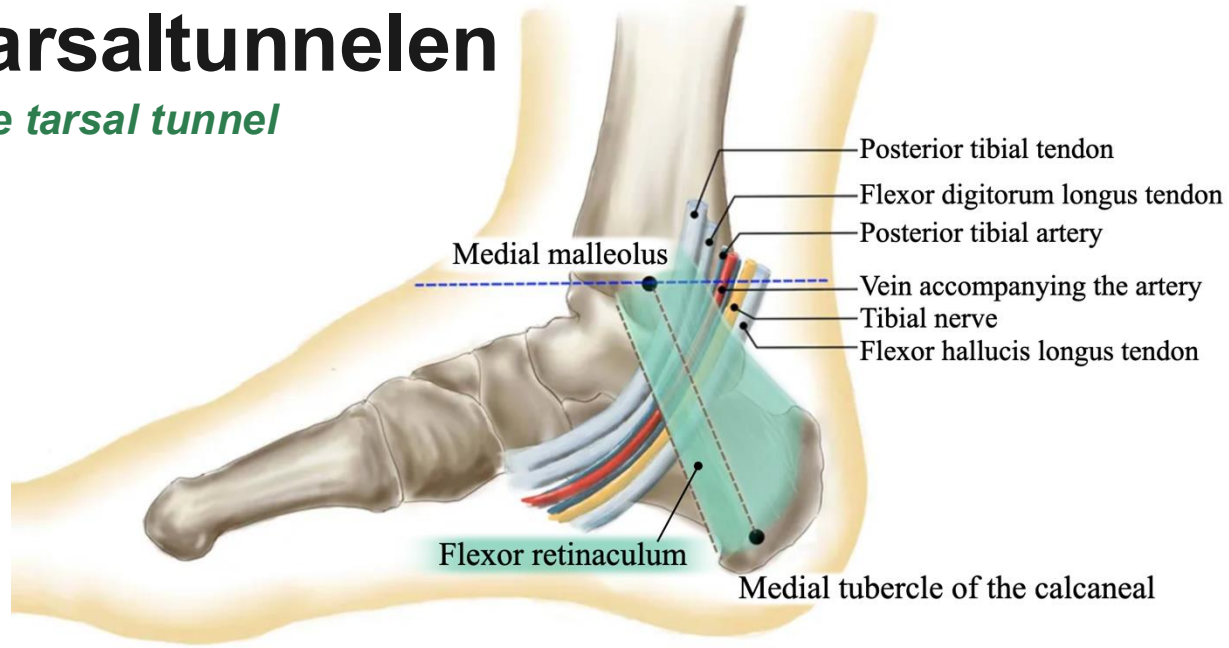
4. **Redegøre for de vigtigste regionale relationer i regio cruris, herunder de fire muskelloger samt forløbet af kar og perifere nerver**
   Describe the major regional relationships in the leg, including the four compartments and the course of vessels and peripheral nerves

5. **Redegøre for underekstremitetens arterielle forsyning, venøse og lymfatiske drænage samt kutane innervation (dermatomer)**
   Describe the arterial supply, venous and lymphatic drainage and cutaneous innervation (dermatomes) of the lower limb

6. Redegøre for de vigtigste passager i regio tarsalis og pedis, herunder tarsaltunnelen, retinacula og planta pedis
   Describe the major passageways of the ankle and foot, including the tarsal tunnel, retinacula and the sole of the foot

Tarsaltunnelen

The tarsal tunnel



Retinaculum flexorum spænder mellem malleolus medialis og calcaneus

The flexor retinaculum spans from the medial malleolus to the calcaneus

Klinik: Tarsaltunnelsyndrom: kompression af n. tibialis giver brændende smerte og paræstesi i fodsålen. A. tibialis posterior kan palperes bag malleolus medialis.

Tarsal tunnel syndrome: compression of the tibial nerve causes burning pain and paraesthesia in the sole. The posterior tibial pulse is palpable behind the medial malleolus.

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Fakta

- Tarsal tunnel syndrom er en smertefuld afklemning af nerven bag ved fodleddets inderste ankelkno
- Den kan skyldes idræt, diabetes eller betændelse ved sener, der medfører tryk på nerven
- Platfod øger også risikoen
- Føleforstyrrelsen går ned i fodsålen, da nerven ligger sammen med sener og kar i en tunnel dækket af et bindevævsbånd henover knoglen
- Man går typisk på ydersiden af foden for at lindre smerten
- Behandlingen kan være indsprøjtning af blokade eller eventuelt en operation med delvis løsning af bindevævsbåndet

Hvordan stilles diagnosen?

Lægen stiller normalt diagnosen ved at undersøge fodleddet og trykke på nerven, hvor der så kommer smerter, der stråler ud i foden. Det giver en fornemmelse af elektrisk stød.

Når foden løftes opad, og fodsålen vendes udad, kommer der ekstra pres i tunnelen. Trykket på nerven stiger derved, og smerten og føleforstyrrelsen bliver værre.

Undersøgelse af nerveledningshastigheden i nerven er også en mulighed, hvor man kan se reduceret udslag af impulser.

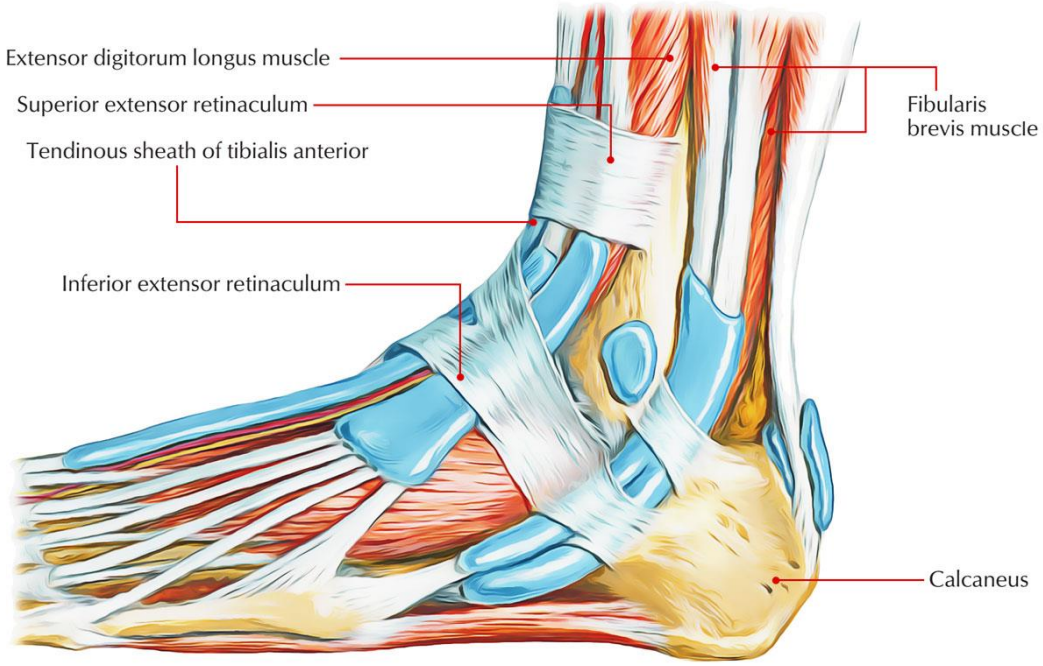
Røntgenundersøgelse giver svar på forholdene omkring knoglerne. Derved kan man vurdere, om knoglenydannelse er årsag til trykgenerne.

Der kan være brug for at supplere med ultralydsskanning, CT-skanning eller MR-skanning for at finde årsagen til sygdommen.

Retinacula og sener omkring anklen

Retinacula and tendons around the ankle

Region *Region*	**Retinaculum**	**Strukturer**
Foran *Anterior*	*Retinaculum extensorum superius* og *inferius*	*M. tibialis anterior*, *m. extensor hallucis longus*, *a. og v. dorsalis pedis* og *n. fibularis profundus*, *m. extensor digitorum longus*, *m. fibularis tertius*
Medialt *Medial*	*Retinaculum flexorum*	Tarsaltunnelens indhold. Fra anterior til posterior: tibialis posterior tendon, flexor digitorum longus tendon, posterior tibial artery and vein, tibial nerve, flexor hallucis longus tendon
Lateralt *Lateral*	*Retinacula mm. fibularium*	*M. fibularis longus* og *brevis*; *n. suralis* passerer bag malleolen



Klinik: Retinacula holder senerne på plads. A. dorsalis pedis-pulsen mærkes lateralt for senen fra m. extensor hallucis longus (mangler hos ca. 10 % af raske). Under retinaculum flexorum kan n. tibialis komprimeres (tarsaltunnelsyndrom)

Retinacula hold the tendons in place. The dorsalis pedis pulse is felt lateral to the extensor hallucis longus tendon (absent in about 10% of healthy people). The tibial nerve can be compressed under the flexor retinaculum (tarsal tunnel syndrome).

Planta pedis

The sole of the foot

Aponeurosis plantaris

Plantar aponeurosis

Fra *tuber calcanei* til tæerne

From the heel to the toes

Støtter fodbuen

Supports the arch

Fire lag

Four layers

Lag 1–4 af korte fodmuskler

Layers 1–4 of intrinsic muscles

Lange sener: tibialis posterior og fibularis longus

Long tendons

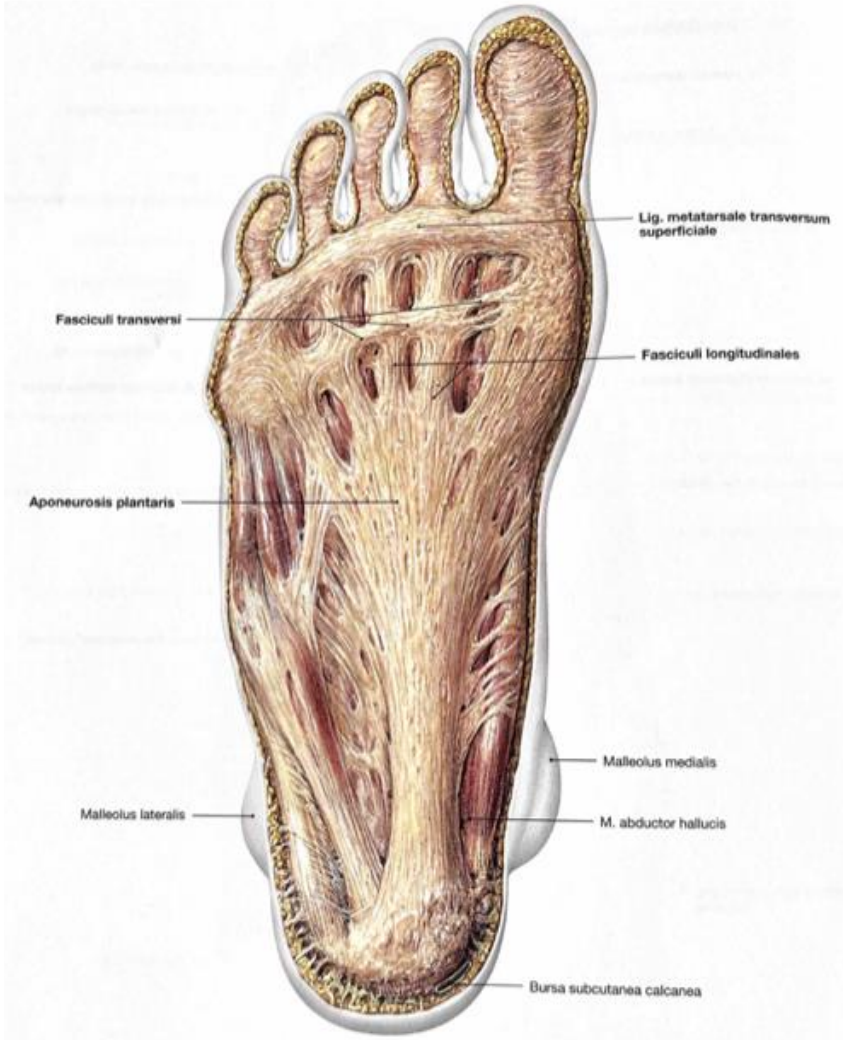
Nerver og kar

Nerves and vessels

N. plantaris medialis og *lateralis*

Arcus plantaris

Plantar arterial arch



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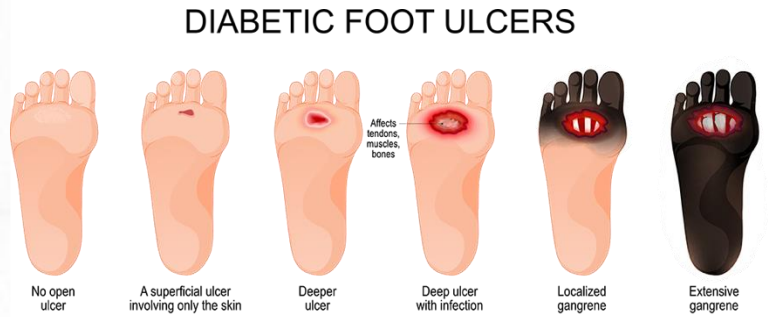
Definition af diabetisk fodsår

- Enhver huddefekt hos en patient med diabetes

Forebyggelse af diabetisk fodsår

Årlig kontrol

- Alle patienter med diabetes bør have undersøgt fødderne mindst én gang om året - kan foretages af læge, sygeplejerske eller fodterapeut
- Undersøgelsen omfatter screening for risikofaktorerne neuropati, arteriosklerose og deformiteter
- Resultatet af denne undersøgelse danner grundlag for risikovurdering bl.a. ved stratificering af behandling ved fodterapeut
- Alle patienter bør tilbydes henvisning til fodterapeut mhp. fodterapi, vejledning i fodhygiejne, valg af fodtøj mm. Der ydes tilskud til fodterapi afhængig af komplikationer som beskrevet



Klinik: Diabetisk fodsår: neuropati (tab af beskyttende følesans, atrofi af fodens korte muskler), forhøjet tryk og ofte nedsat arteriel cirkulation

Diabetic foot ulcer: neuropathy (loss of protective sensation, atrophy of the intrinsic foot muscles), raised pressure, and often impaired arterial circulation.

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Litteratur og læsning

Reading

Bevægeapparatets anatomi (modulets danske lærebog)
The Danish textbook for the module

Sobotta Atlas of Anatomy: Underekstremitet
Lower limb volume

Muskelskema (modulets materiale på Moodle)
Muscle table, course material